

[Day 8 종합실습-4] ecommerce 매출 분석 및 성능 튜닝

AI 서비스를 위한 SW 기초 Full-stack Engineering — 스마트 데이터 이해 및 활용 | 종합실습-4 | 광주캠퍼스 4반 박영서 | 2026-07-24

실습 결과 요약

- 실행환경: PostgreSQL 17.10 — DB skala_db · 스키마 ecom · EXPLAIN (ANALYZE, BUFFERS)로 계획·수행시간 검증
- 데이터: customers 3,000 · products 600 · orders 6,770 · order_items 18,393 · reviews 2,064 · inventory 600
- 제출물 1: Q1~Q11 각 문항 튜닝 전/후 쿼리·실행계획·결과 (병목 → 튜닝 기법 → 개선 검증)
- 제출물 2: 3가지 조인 알고리즘(Nested Loop / Hash / Merge) 실행계획·특성 비교
- 제출물 3: Materialized View(mv_daily_gmv) 생성·갱신·리포트 가속(약 860배) 및 갱신 전략
- 튜닝 기법: 부분·복합·커버링 인덱스, 쿼리 재작성(count(DISTINCT) 제거·불필요 Sort 제거), 비용기반 옵티마이저 크로스오버 분석
- 채점 기준 준수 — 요구사항 1:1 반영 · 필요 컬럼만 선택(SELECT * 미사용) · DB 프로그래밍 주석 처리(모든 SQL 파일에 병목·튜닝 근거 주석)

측정 방법 및 유의점 — 본 데이터는 소규모(orders 6,770·order_items 18,393)라 절대 실행시간이 수 ms대이며, EXPLAIN ANALYZE의 행별 계측 오버헤드로 단일 실행 시간은 실행마다 편차가 크다(같은 쿼리도 run마다 수 ms 출력임). 따라서 각 문항의 튜닝 효과는 실행계획 노드(스캔·조인·정렬)·버퍼·접근 행수의 변화를 핵심 근거로 판단하였고(EXPLAIN ANALYZE, BUFFERS), 종합 요약 표의 실행시간은 각 문항 캡처의 Execution Time(전→후)을 그대로 옮긴 참고값이다. 이 이득은 데이터가 커질수록 확대된다.

0. 실습 환경 확인

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x43
skala_db=# \i env.sql
\conninfo
접속 정보: 데이터베이스="skala_db", 사용자="parkyoungseo", 호스트="localhost" (주소="::1"), 포트="5432".
\dt ecom.*
          테이블명          | 테이블 | 소유주
-----|-----|-----
 ecom | addresses | parkyoungseo
 ecom | categories | parkyoungseo
 ecom | country   | parkyoungseo
 ecom | customers | parkyoungseo
 ecom | inventory | parkyoungseo
 ecom | order_items | parkyoungseo
 ecom | orders    | parkyoungseo
 ecom | payments  | parkyoungseo
 ecom | product_prices | parkyoungseo
 ecom | product_suppliers | parkyoungseo
 ecom | products  | parkyoungseo
 ecom | reviews   | parkyoungseo
 ecom | shipments | parkyoungseo
 ecom | suppliers | parkyoungseo
(14개 행)

SELECT 'orders' t,count(*) FROM orders UNION ALL SELECT 'order_items',count(*) FROM order_items
UNION ALL SELECT 'customers',count(*) FROM customers UNION ALL SELECT 'products',count(*) FROM products
UNION ALL SELECT 'reviews',count(*) FROM reviews UNION ALL SELECT 'inventory',count(*) FROM inventory ORDER BY 1;
 t      | count
-----|-----
customers | 3000
inventory  | 600
order_items | 18393
orders     | 6770
products   | 600
reviews    | 2064
(6개 행)

skala_db=#
```

▲ skala_db / ecom 스키마 접속(\conninfo), 테이블 목록·행 수 확인

제출물 1 · Q1~Q11 튜닝 전/후 비교

Q1. 지난 한 달간 실제 판매 총액 (paid/shipped/delivered) 인덱스 추가

요구사항 — 취소·환불·생성 상태 제외, 최근 1개월(now 기준) 매출 합계. 선택 컬럼: revenue 1개.

병목 — orders에 (상태 AND 기간) 필터가 있으나 이를 지원하는 인덱스가 없어 Seq Scan on orders(6,770행 스캔 → 5,339행 필터 제거).

튜닝: 부분 인덱스 — 매출 상태만 담고 order_ts로 정렬

SQL — 튜닝 전 쿼리

```
SELECT sum(oi.line_total) AS revenue
FROM   orders o
JOIN    order_items oi ON oi.order_id = o.order_id
WHERE   o.order_status IN ('paid','shipped','delivered')
AND     o.order_ts >= now() - interval '1 month';
```

-- 총매출 = 상세금액 전부 합산
-- 주문 헤더(상태/기간 필터 대상)
-- 주문 1건에 딸린 상세 라인 연결
-- 매출 인정 상태만(취소/환불 제외)
-- 최근 1개월로 기간 한정

SQL — 튜닝 후 (인덱스 DDL)

```
-- 부분 인덱스만 추가, 쿼리는 튜닝 전과 동일
CREATE INDEX idx_orders_rev_ts ON orders(order_ts)
WHERE order_status IN ('paid','shipped','delivered');
```

-- order_ts를 정렬키로
-- 매출 상태만 인덱싱(부분 인덱스)

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x34

skala_db=# \i q01_before.sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT sum(oi.line_total) AS revenue
FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
WHERE o.order_status IN ('paid','shipped','delivered')
AND o.order_ts >= now() - interval '1 month';

QUERY PLAN

Aggregate (cost=651.39..651.40 rows=1 width=32) (actual time=8.493..8.498 rows=1 loops=1)
  Buffers: shared hit=248
  -> Hash Join (cost=231.54..641.78 rows=3842 width=6) (actual time=2.852..7.949 rows=3965 loops=1)
    Hash Cond: (oi.order_id = o.order_id)
    Buffers: shared hit=248
    -> Seq Scan on order_items oi (cost=0.00..361.93 rows=18393 width=14) (actual time=0.813..1.974 rows=18393 loops=1)
      Buffers: shared hit=178
    -> Hash (cost=213.86..213.86 rows=1414 width=8) (actual time=2.785..2.786 rows=1431 loops=1)
      Buckets: 2048 Batches: 1 Memory Usage: 72kB
      Buffers: shared hit=70
      -> Seq Scan on orders o (cost=0.00..213.86 rows=1414 width=8) (actual time=0.819..2.458 rows=1431 loops=1)
        Filter: ((order_status = ANY ('{paid,shipped,delivered}'::text[])) AND (order_ts >= (now() - '1 mon'::interval)))
        Rows Removed by Filter: 5339
        Buffers: shared hit=70
Planning:
  Buffers: shared hit=45
Planning Time: 1.384 ms
Execution Time: 8.578 ms
(18개 행)

skala_db=#
```

▲ 튜닝 전 — EXPLAIN (ANALYZE, BUFFERS) baseline 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x38

skala_db=# \i q01_after.sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT sum(oi.line_total) AS revenue
FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
WHERE o.order_status IN ('paid','shipped','delivered')
AND o.order_ts >= now() - interval '1 month';

QUERY PLAN

Aggregate (cost=568.82..568.83 rows=1 width=32) (actual time=7.449..7.450 rows=1 loops=1)
  Buffers: shared hit=254
  -> Hash Join (cost=148.97..559.21 rows=3842 width=6) (actual time=0.841..6.845 rows=3965 loops=1)
    Hash Cond: (oi.order_id = o.order_id)
    Buffers: shared hit=254
    -> Seq Scan on order_items oi (cost=0.00..361.93 rows=18393 width=14) (actual time=0.812..2.453 rows=18393 loops=1)
      Buffers: shared hit=178
    -> Hash (cost=131.29..131.29 rows=1414 width=8) (actual time=0.812..0.813 rows=1431 loops=1)
      Buckets: 2048 Batches: 1 Memory Usage: 72kB
      Buffers: shared hit=76
      -> Bitmap Heap Scan on orders o (cost=31.25..131.29 rows=1414 width=8) (actual time=0.127..0.604 rows=1431 loops=1)
        Recheck Cond: ((order_ts >= (now() - '1 mon'::interval)) AND (order_status = ANY ('{paid,shipped,delivered}'::text[])))
        Heap Blocks: exact=70
        Buffers: shared hit=76
        -> Bitmap Index Scan on idx_orders_rev_ts (cost=0.00..30.89 rows=1414 width=0) (actual time=0.100..0.100 rows=1431 loops=1)
          Index Cond: (order_ts >= (now() - '1 mon'::interval))
          Buffers: shared hit=6
Planning:
  Buffers: shared hit=92
Planning Time: 1.836 ms
Execution Time: 7.490 ms
(21개 행)

skala_db=#
```

▲ 튜닝 후 — 인덱스/재작성 적용 후 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x15

skala_db=# \i q01_result.sql
SELECT sum(oi.line_total) AS revenue
FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
WHERE o.order_status IN ('paid','shipped','delivered')
AND o.order_ts >= now() - interval '1 month';
revenue
1395890.20
(1개 행)

skala_db=#
```

▲ 실행 결과 (튜닝 전후 동일 — 정확성 보존)

개선 — orders 접근이 Seq Scan(6,770행 전수) → Bitmap Index Scan (idx_orders_rev_ts, 1,431행). 필터 테이블을 인덱스로 직접 접근. 총 실행시간은 order_items 공유 스캔이 지배하여 유사하나, 데이터가 커질수록 인덱스 이득이 확대.

결과 — revenue = 1,395,893.20 (튜닝 전후 동일)

비즈니스 인사이트 — 지난 한 달 실결제 매출 약 **139만** — 취소·환불·생성을 제외한 '실제 확정 매출'만 집계하여 재무 마감·실적 지표로 바로 활용 가능.

Q2. 월별 주문 수 / 매출 / 주문당 평균금액(AOV) 쿼리 재작성

요구사항 — 월별 주문 건수·매출 합계·AOV(매출/주문수). 선택 컬럼: mon, orders, revenue, aov.

병목 — AOV용 count(DISTINCT order_id)가 GroupAggregate 입력을 (month, order_id)로 정렬 → 14,250행 Sort(quicksort 941kB).

튜닝: 쿼리 재작성 — 주문 단위 선집계 후 월별 재집계 (count(DISTINCT) 제거)

SQL — 튜닝 전 쿼리

```
SELECT date_trunc('month', o.order_ts) AS mon,          -- 월 단위 절삭(월별 그룹키)
       count(DISTINCT o.order_id) AS orders,           -- 주문수: 조인 중복 제거 위해 DISTINCT
       sum(oi.line_total) AS revenue,                 -- 월 매출 합계
       sum(oi.line_total)/count(DISTINCT o.order_id) AS aov -- AOV = 매출 / 주문수
FROM   orders o
JOIN   order_items oi ON oi.order_id = o.order_id
WHERE  o.order_status IN ('paid','shipped','delivered') -- 매출 인정 상태만
GROUP BY 1 ORDER BY 1;                                -- 월별로 그룹·정렬
```

SQL — 튜닝 후 (재작성 쿼리)

```
-- 주문 단위 선집계 후 월별 재집계 (count(DISTINCT) 제거)
SELECT date_trunc('month', order_ts) AS mon, count(*) AS orders, -- 이미 주문단위 → count(*)
       sum(order_rev) AS revenue, round(sum(order_rev)/count(*),2) AS aov
FROM (
  SELECT o.order_id, o.order_ts, sum(oi.line_total) AS order_rev -- 주문별 금액 선집계
  FROM   orders o JOIN order_items oi ON oi.order_id = o.order_id
  WHERE  o.order_status IN ('paid','shipped','delivered')
  GROUP BY o.order_id, o.order_ts
) t
GROUP BY 1 ORDER BY 1;                                -- 월별 재집계 (HashAggregate)
```

```
parkyoungseo - psql -h localhost -U parkyoungseo -d skala_db - 183x43
skala_db=# \i q02_before.sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT date_trunc('month', o.order_ts) AS mon,
       count(DISTINCT o.order_id) AS orders,
       sum(oi.line_total) AS revenue,
       sum(oi.line_total)/count(DISTINCT o.order_id) AS aov
FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
WHERE o.order_status IN ('paid','shipped','delivered')
GROUP BY 1 ORDER BY 1;

QUERY PLAN

GroupAggregate (cost=1651.10..1897.13 rows=5216 width=80) (actual time=18.521..21.101 rows=6 loops=1)
  Group Key: (date_trunc('month'::text, o.order_ts))
  Buffers: shared hit=251
  -> Sort (cost=1651.10..1686.52 rows=14171 width=22) (actual time=18.432..19.114 rows=14250 loops=1)
    Sort Key: (date_trunc('month'::text, o.order_ts)), o.order_id
    Sort Method: quicksort Memory: 941kB
    Buffers: shared hit=251
    -> Hash Join (cost=228.29..673.96 rows=14171 width=22) (actual time=3.625..13.075 rows=14250 loops=1)
      Hash Cond: (oi.order_id = o.order_id)
      Buffers: shared hit=248
      -> Seq Scan on order_items oi (cost=0.00..361.93 rows=18393 width=14) (actual time=0.004..1.641 rows=18393 loops=1)
        Buffers: shared hit=178
      -> Hash (cost=163.09..163.09 rows=5216 width=16) (actual time=3.603..3.603 rows=5216 loops=1)
        Buckets: 8192 Batches: 1 Memory Usage: 309kB
        Buffers: shared hit=70
        -> Seq Scan on orders o (cost=0.00..163.09 rows=5216 width=16) (actual time=0.004..2.636 rows=5216 loops=1)
          Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
          Rows Removed by Filter: 1554
          Buffers: shared hit=70
Planning:
  Buffers: shared hit=22
Planning Time: 0.200 ms
Execution Time: 21.134 ms
(23개 행)

skala_db=#
```

▲ 튜닝 전 — EXPLAIN (ANALYZE, BUFFERS) baseline 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x51
skala_db=# \i q02_after.sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT date_trunc('month', order_ts) AS mon, count(*) AS orders,
       sum(order_rev) AS revenue, round(sum(order_rev)/count(*),2) AS aov
FROM (SELECT o.order_id, o.order_ts, sum(oi.line_total) AS order_rev
      FROM orders o JOIN order_items oi ON oi.order_id=o.order_id
      WHERE o.order_status IN ('paid','shipped','delivered')
      GROUP BY o.order_id, o.order_ts) t
GROUP BY 1 ORDER BY 1;

QUERY PLAN

Sort  (cost=891.06..891.55 rows=200 width=80) (actual time=19.711..19.714 rows=6 loops=1)
  Sort Key: (date_trunc('month'::text, t.order_ts))
  Sort Method: quicksort  Memory: 25kB
  Buffers: shared hit=248
  -> HashAggregate  (cost=878.91..883.41 rows=200 width=80) (actual time=19.701..19.707 rows=6 loops=1)
    Group Key: date_trunc('month'::text, t.order_ts)
    Batches: 1  Memory Usage: 40kB
    Buffers: shared hit=248
    -> Subquery Scan on t  (cost=709.39..839.79 rows=5216 width=48) (actual time=15.661..18.741 rows=5216 loops=1)
      Buffers: shared hit=248
      -> HashAggregate  (cost=709.39..774.59 rows=5216 width=48) (actual time=15.655..17.360 rows=5216 loops=1)
        Group Key: o.order_id
        Batches: 1  Memory Usage: 2257kB
        Buffers: shared hit=248
        -> Hash Join  (cost=228.29..638.53 rows=14171 width=22) (actual time=2.837..10.172 rows=14250 loops=1)
          Hash Cond: (oi.order_id = o.order_id)
          Buffers: shared hit=248
          -> Seq Scan on order_items oi  (cost=0.00..361.93 rows=18393 width=14) (actual time=0.004..1.772 rows=18393 loops=1)
            Buffers: shared hit=178
          -> Hash  (cost=163.09..163.09 rows=5216 width=16) (actual time=2.824..2.824 rows=5216 loops=1)
            Buckets: 8192  Batches: 1  Memory Usage: 389kB
            Buffers: shared hit=70
            -> Seq Scan on orders o  (cost=0.00..163.09 rows=5216 width=16) (actual time=0.004..2.866 rows=5216 loops=1)
              Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
              Rows Removed by Filter: 1554
              Buffers: shared hit=70

Planning:
  Buffers: shared hit=14
Planning Time: 0.478 ms
Execution Time: 19.760 ms
(30개 행 )

skala_db=#
```

▲ 튜닝 후 — 인덱스/재작성 적용 후 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x23
skala_db=# \i q02_result.sql
SELECT date_trunc('month', order_ts)::date AS mon, count(*) AS orders,
       sum(order_rev) AS revenue, round(sum(order_rev)/count(*),2) AS aov
FROM (SELECT o.order_id, o.order_ts, sum(oi.line_total) AS order_rev
      FROM orders o JOIN order_items oi ON oi.order_id=o.order_id
      WHERE o.order_status IN ('paid','shipped','delivered')
      GROUP BY o.order_id, o.order_ts) t
GROUP BY 1 ORDER BY 1;

 mon | orders | revenue | aov
-----+-----+-----+----
2026-03-01 | 178 | 158853.58 | 847.49
2026-04-01 | 1199 | 1877658.71 | 985.59
2026-05-01 | 1523 | 1183546.21 | 889.22
2026-06-01 | 1478 | 1397944.57 | 945.85
2026-07-01 | 1115 | 1889484.21 | 977.12
2026-08-01 | 2 | 788.18 | 394.09
(6개 행 )

skala_db=#
```

▲ 실행 결과 (튜닝 전후 동일 — 정확성 보존)

개선 — GroupAggregate + Sort(14,250행 정렬, 941kB) → 2단 HashAggregate. count(DISTINCT)가 유발하던 대형 정렬 노드가 계획에서 사라짐.

결과 — 4~7월 월매출 108만~140만, AOV 880~977 (전후 동일)

비즈니스 인사이트 — 월 매출은 6월(약 140만)에 정점, AOV가 880→977로 완만한 우상향 — 객단가 개선 추세(3·8월은 데이터 경계라 건수 적음).

Q3. 최근 90일 카테고리 Top10 (매출 기준) 계획 분석:크로스오버

요구사항 — 최근 90일, 카테고리별 매출 합계 상위 10. 선택 컬럼: category_id, category_name, revenue.

병목 — orders를 (상태 AND 90일) 필터로 Seq Scan 후 order_items·products·categories 4중 JOIN.

튜닝: 부분 인덱스(idx_orders_rev_ts) 활용 검토 — 단, 90일은 전체의 약 60%로 저선택도

SQL — 튜닝 전 쿼리

```
SELECT c.category_id, c.category_name, sum(oi.line_total) AS revenue -- 카테고리별 매출
FROM   orders o
JOIN   order_items oi ON oi.order_id = o.order_id -- 주문 → 상세
JOIN   products p ON p.product_id = oi.product_id -- 상세 → 상품
JOIN   categories c ON c.category_id = p.category_id -- 상품 → 카테고리
WHERE  o.order_status IN ('paid','shipped','delivered') -- 매출 인정 상태만
      AND o.order_ts >= now() - interval '90 days' -- 최근 90일
GROUP BY c.category_id, c.category_name
ORDER BY revenue DESC LIMIT 10; -- 매출 상위 10
```

튜닝 후 (계획 분석 — 쿼리 동일)

-- 90일 ≈ 60% 저선택도 → 옵티마이저가 Seq Scan 유지가 최적(강제 시 악화).
-- 부분 인덱스는 존재하나 이 조건에선 미채택. 쿼리 동일. (통계 최신화: ANALYZE)
-- 검증: SET enable_seqscan=off → Bitmap Index Scan(idx_orders_rev_ts) 동작 확인

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x65

skala_db=# \i q03_before.sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT c.category_id, c.category_name, sum(oi.line_total) AS revenue
FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
JOIN products p ON p.product_id = oi.product_id
JOIN categories c ON c.category_id = p.category_id
WHERE o.order_status IN ('paid','shipped','delivered')
      AND o.order_ts >= now() - interval '90 days'
GROUP BY c.category_id, c.category_name
ORDER BY revenue DESC LIMIT 10;

QUERY PLAN

-----
Limit (cost=877.61..877.63 rows=10 width=72) (actual time=12.549..12.556 rows=10 loops=1)
  Buffers: shared hit=259
  -> Sort (cost=877.61..880.28 rows=1070 width=72) (actual time=12.548..12.553 rows=10 loops=1)
    Sort Key: (sum(oi.line_total)) DESC
    Sort Method: quicksort Memory: 25kB
    Buffers: shared hit=259
    -> HashAggregate (cost=841.11..854.48 rows=1070 width=72) (actual time=12.514..12.526 rows=10 loops=1)
      Group Key: c.category_id
      Batches: 1 Memory Usage: 73kB
      Buffers: shared hit=256
      -> Hash Join (cost=318.66..786.53 rows=10916 width=46) (actual time=3.366..10.658 rows=11106 loops=1)
        Hash Cond: (p.category_id = c.category_id)
        Buffers: shared hit=256
        -> Hash Join (cost=284.59..723.69 rows=10916 width=14) (actual time=3.346..9.198 rows=11106 loops=1)
          Hash Cond: (oi.product_id = p.product_id)
          Buffers: shared hit=255
          -> Hash Join (cost=264.09..674.33 rows=10916 width=14) (actual time=3.227..7.521 rows=11106 loops=1)
            Hash Cond: (oi.order_id = o.order_id)
            Buffers: shared hit=248
            -> Seq Scan on order_items oi (cost=0.00..361.93 rows=18393 width=22) (actual time=0.003..1.015 rows=18393 loops=1)
              Buffers: shared hit=178
            -> Hash (cost=213.86..213.86 rows=4018 width=8) (actual time=3.212..3.213 rows=4056 loops=1)
              Buckets: 4096 Batches: 1 Memory Usage: 191kB
              Buffers: shared hit=70
              -> Seq Scan on orders o (cost=0.00..213.86 rows=4018 width=8) (actual time=0.006..2.767 rows=4056 loops=1)
                Filter: ((order_status = ANY ('paid,shipped,delivered')::text[])) AND (order_ts >= (now() - '90 days'::interval)))
                Rows Removed by Filter: 2714
                Buffers: shared hit=70
          -> Hash (cost=13.00..13.00 rows=600 width=16) (actual time=0.099..0.100 rows=600 loops=1)
            Buckets: 1024 Batches: 1 Memory Usage: 37kB
            Buffers: shared hit=7
            -> Seq Scan on products p (cost=0.00..13.00 rows=600 width=16) (actual time=0.003..0.053 rows=600 loops=1)
              Buffers: shared hit=7
        -> Hash (cost=20.70..20.70 rows=1070 width=40) (actual time=0.012..0.013 rows=14 loops=1)
          Buckets: 2048 Batches: 1 Memory Usage: 17kB
          Buffers: shared hit=1
          -> Seq Scan on categories c (cost=0.00..20.70 rows=1070 width=40) (actual time=0.007..0.009 rows=14 loops=1)
            Buffers: shared hit=1

Planning:
  Buffers: shared hit=150
  Planning Time: 1.074 ms
  Execution Time: 12.657 ms
(42개 행)

skala_db=#
```

▲ 튜닝 전 — EXPLAIN (ANALYZE, BUFFERS) baseline 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x65

skala_db=# \i q03_after.sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT c.category_id, c.category_name, sum(oi.line_total) AS revenue
FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
JOIN products p ON p.product_id = oi.product_id
JOIN categories c ON c.category_id = p.category_id
WHERE o.order_status IN ('paid','shipped','delivered')
AND o.order_ts >= now() - interval '90 days'
GROUP BY c.category_id, c.category_name
ORDER BY revenue DESC LIMIT 10;

QUERY PLAN

Limit (cost=877.61..877.63 rows=10 width=72) (actual time=13.838..13.843 rows=10 loops=1)
  Buffers: shared hit=256
  -> Sort (cost=877.61..880.28 rows=1070 width=72) (actual time=13.837..13.840 rows=10 loops=1)
    Sort Key: (sum(oi.line_total)) DESC
    Sort Method: quicksort Memory: 25kB
    Buffers: shared hit=256
    -> HashAggregate (cost=841.11..854.48 rows=1070 width=72) (actual time=13.818..13.831 rows=10 loops=1)
      Group Key: c.category_id
      Batches: 1 Memory Usage: 73kB
      Buffers: shared hit=256
      -> Hash Join (cost=318.66..786.53 rows=10916 width=46) (actual time=3.506..11.775 rows=11186 loops=1)
        Hash Cond: (p.category_id = c.category_id)
        Buffers: shared hit=256
        -> Hash Join (cost=284.59..723.69 rows=10916 width=14) (actual time=3.491..9.936 rows=11186 loops=1)
          Hash Cond: (oi.product_id = p.product_id)
          Buffers: shared hit=255
          -> Hash Join (cost=264.09..674.33 rows=10916 width=14) (actual time=3.296..7.794 rows=11186 loops=1)
            Hash Cond: (oi.order_id = o.order_id)
            Buffers: shared hit=248
            -> Seq Scan on order_items oi (cost=0.00..361.93 rows=18393 width=22) (actual time=0.004..1.313 rows=18393 loops=1)
              Buffers: shared hit=178
            -> Hash (cost=213.86..213.86 rows=4018 width=8) (actual time=3.278..3.279 rows=4056 loops=1)
              Buckets: 4096 Batches: 1 Memory Usage: 191kB
              Buffers: shared hit=70
              -> Seq Scan on orders o (cost=0.00..213.86 rows=4018 width=8) (actual time=0.014..2.215 rows=4056 loops=1)
                Filter: ((order_status = ANY ('{paid,shipped,delivered}'::text[])) AND (order_ts >= (now() - '90 days'::interval)))
                Rows Removed by Filter: 2714
                Buffers: shared hit=70
          -> Hash (cost=13.00..13.00 rows=600 width=16) (actual time=0.178..0.178 rows=600 loops=1)
            Buckets: 1024 Batches: 1 Memory Usage: 37kB
            Buffers: shared hit=7
            -> Seq Scan on products p (cost=0.00..13.00 rows=600 width=16) (actual time=0.003..0.121 rows=600 loops=1)
              Buffers: shared hit=7
        -> Hash (cost=20.70..20.70 rows=1070 width=40) (actual time=0.011..0.011 rows=14 loops=1)
          Buckets: 2048 Batches: 1 Memory Usage: 17kB
          Buffers: shared hit=1
          -> Seq Scan on categories c (cost=0.00..20.70 rows=1070 width=40) (actual time=0.007..0.008 rows=14 loops=1)
            Buffers: shared hit=1

Planning:
  Buffers: shared hit=29
Planning Time: 0.775 ms
Execution Time: 13.884 ms
(42개 행 )

skala_db=# 프
```

▲ 튜닝 후 — 인덱스/재작성 적용 후 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x29

skala_db=# \i q03_result.sql
SELECT c.category_id, c.category_name, sum(oi.line_total) AS revenue
FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
JOIN products p ON p.product_id = oi.product_id
JOIN categories c ON c.category_id = p.category_id
WHERE o.order_status IN ('paid','shipped','delivered')
AND o.order_ts >= now() - interval '90 days'
GROUP BY c.category_id, c.category_name
ORDER BY revenue DESC LIMIT 10;

category_id | category_name | revenue
-----
10 | Women | 1328413.25
11 | Shoes | 881934.86
16 | Fitness | 212921.92
2 | Phones | 204866.51
4 | Audio | 204866.32
9 | Men | 198227.47
13 | Outdoor | 193991.90
3 | Laptops | 192368.10
6 | Appliances | 190218.29
7 | Cookware | 184468.98

(10개 행 )

skala_db=#
```

▲ 실행 결과 (튜닝 전후 동일 — 정확성 보존)

개선 — 이 규모(90일 ~ 60%)에선 **Seq Scan 유지가 최적**(옵티마이저 판단 정당). ANALYZE로 categories 추정행 1,070→14 정확화. 강제 인덱스 시 오히려 악화 → 인덱스가 만능이 아님을 실증.

결과 — Women 132만 > Shoes 88만 > Fitness 21만 … (Top10, 전후 동일)

비즈니스 인사이트 — **Women·Shoes가 매출을 견인**(합계 약 220만), 이어 Fitness·Phones·Audio. 상위 2개 카테고리 집중도가 높아 재고·프로모션 우선순위 대상.

Q4. 제품별 누적 매출 RANK() Top20 계획 분석·크로스오버

요구사항 — 제품별 매출 합계에 RANK() 부여, 상위 20. 선택 컬럼: product_id, product_name, revenue, rnk.

병목 — 필터가 상태뿐이라 order_items 전량(18,393) 집계 불가피. RANK()는 정의상 전체 Sort 필요 → 계산 바운드.

튜닝: 커버링 인덱스 검토 — 단, 전량 집계형이라 인덱스로는 스캔 축소 불가

SQL — 튜닝 전 쿼리

```
SELECT product_id, product_name, revenue,
       RANK() OVER (ORDER BY revenue DESC) AS rnk      -- 매출 내림차순 순위 부여
FROM (
  SELECT p.product_id, p.product_name, sum(oi.line_total) AS revenue -- 제품별 매출 선집계
  FROM   order_items oi
  JOIN   orders o ON o.order_id = oi.order_id          -- 상태 필터용 주문 조인
  JOIN   products p ON p.product_id = oi.product_id    -- 상품명 조인
  WHERE  o.order_status IN ('paid','shipped','delivered')
  GROUP BY p.product_id, p.product_name
) t
ORDER BY rnk LIMIT 20;                                -- 순위 상위 20
```

튜닝 후 (계획 분석 — 쿼리 동일)

-- 전량 집계 + RANK() 정렬은 불가피(계산 바운드) → 인덱스로 스캔 축소 불가.
-- 근본 개선책은 사전 집계(재출물 3의 Materialized View). 쿼리 동일.

```
parkyoungseo ~ psql -h localhost -U parkyoungseo -d skala_db - 183x64

skala_db=# \i q04_before.sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT product_id, product_name, revenue, RANK() OVER (ORDER BY revenue DESC) AS rnk
FROM (SELECT p.product_id, p.product_name, sum(oi.line_total) AS revenue
      FROM order_items oi JOIN orders o ON o.order_id=oi.order_id
      JOIN products p ON p.product_id=oi.product_id
      WHERE o.order_status IN ('paid','shipped','delivered')
      GROUP BY p.product_id, p.product_name) t
ORDER BY rnk LIMIT 20;

                                QUERY PLAN

Limit  (cost=829.01..829.06 rows=20 width=59) (actual time=14.163..14.168 rows=20 loops=1)
 Buffers: shared hit=255
-> Sort (cost=829.01..830.51 rows=600 width=59) (actual time=14.162..14.165 rows=20 loops=1)
   Sort Key: (rank() OVER (?))
   Sort Method: top-N heapsort  Memory: 26kB
   Buffers: shared hit=255
-> WindowAgg (cost=802.56..813.04 rows=600 width=59) (actual time=13.872..14.091 rows=600 loops=1)
   Buffers: shared hit=255
   -> Sort (cost=802.54..804.04 rows=600 width=51) (actual time=13.857..13.880 rows=600 loops=1)
     Sort Key: t.revenue DESC
     Sort Method: quicksort  Memory: 53kB
     Buffers: shared hit=255
     -> Subquery Scan on t (cost=767.35..774.85 rows=600 width=51) (actual time=13.524..13.677 rows=600 loops=1)
       Buffers: shared hit=255
       -> HashAggregate (cost=767.35..774.85 rows=600 width=51) (actual time=13.523..13.637 rows=600 loops=1)
         Group Key: p.product_id
         Batches: 1  Memory Usage: 297kB
         Buffers: shared hit=255
         -> Hash Join (cost=248.79..696.50 rows=14171 width=25) (actual time=2.061..10.525 rows=14250 loops=1)
           Hash Cond: (oi.product_id = p.product_id)
           Buffers: shared hit=255
           -> Hash Join (cost=228.29..638.53 rows=14171 width=14) (actual time=1.952..8.099 rows=14250 loops=1)
             Hash Cond: (oi.order_id = o.order_id)
             Buffers: shared hit=248
             -> Seq Scan on order_items oi (cost=0.00..361.93 rows=18393 width=22) (actual time=0.002..1.187 rows=18393 loops=1)
               Buffers: shared hit=170
             -> Hash (cost=163.09..163.09 rows=5216 width=8) (actual time=1.914..1.914 rows=5216 loops=1)
               Buckets: 8192  Batches: 1  Memory Usage: 268kB
               Buffers: shared hit=70
               -> Seq Scan on orders o (cost=0.00..163.09 rows=5216 width=8) (actual time=0.005..0.931 rows=5216 loops=1)
                 Filter: (order_status = ANY ('paid,shipped,delivered')::text[])
                 Rows Removed by Filter: 1554
                 Buffers: shared hit=70
           -> Hash (cost=13.00..13.00 rows=600 width=19) (actual time=0.102..0.102 rows=600 loops=1)
             Buckets: 1024  Batches: 1  Memory Usage: 39kB
             Buffers: shared hit=7
             -> Seq Scan on products p (cost=0.00..13.00 rows=600 width=19) (actual time=0.006..0.051 rows=600 loops=1)
               Buffers: shared hit=7

Planning:
 Buffers: shared hit=29
Planning Time: 0.339 ms
Execution Time: 14.228 ms
(42개 행)

skala_db=#
```

▲ 튜닝 전 — EXPLAIN (ANALYZE, BUFFERS) baseline 실행계획


```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x64

skala_db=# \i q04_after.sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT product_id, product_name, revenue, RANK() OVER (ORDER BY revenue DESC) AS rnk
FROM (SELECT p.product_id, p.product_name, sum(oi.line_total) AS revenue
      FROM order_items oi JOIN orders o ON o.order_id=oi.order_id
      JOIN products p ON p.product_id=oi.product_id
      WHERE o.order_status IN ('paid','shipped','delivered')
      GROUP BY p.product_id, p.product_name) t
ORDER BY rnk LIMIT 20;

QUERY PLAN

Limit (cost=829.01..829.06 rows=20 width=59) (actual time=18.493..18.497 rows=20 loops=1)
  Buffers: shared hit=255
  -> Sort (cost=829.01..830.51 rows=600 width=59) (actual time=18.492..18.495 rows=20 loops=1)
    Sort Key: (rank() OVER (?))
    Sort Method: top-N heapsort  Memory: 26kB
    Buffers: shared hit=255
    -> WindowAgg (cost=802.56..813.04 rows=600 width=59) (actual time=18.115..18.370 rows=600 loops=1)
      Buffers: shared hit=255
      -> Sort (cost=802.54..804.04 rows=600 width=51) (actual time=18.111..18.139 rows=600 loops=1)
        Sort Key: t.revenue DESC
        Sort Method: quicksort  Memory: 53kB
        Buffers: shared hit=255
        -> Subquery Scan on t (cost=767.35..774.85 rows=600 width=51) (actual time=17.869..18.003 rows=600 loops=1)
          Buffers: shared hit=255
          -> HashAggregate (cost=767.35..774.85 rows=600 width=51) (actual time=17.869..17.969 rows=600 loops=1)
            Group Key: p.product_id
            Batches: 1  Memory Usage: 297kB
            Buffers: shared hit=255
            -> Hash Join (cost=248.79..696.50 rows=14171 width=25) (actual time=1.797..13.336 rows=14250 loops=1)
              Hash Cond: (oi.product_id = p.product_id)
              Buffers: shared hit=255
              -> Hash Join (cost=228.29..638.53 rows=14171 width=14) (actual time=1.691..9.002 rows=14250 loops=1)
                Hash Cond: (oi.order_id = o.order_id)
                Buffers: shared hit=248
                -> Seq Scan on order_items oi (cost=0.00..361.93 rows=18393 width=22) (actual time=0.002..2.242 rows=18393 loops=1)
                  Buffers: shared hit=178
                  -> Hash (cost=163.09..163.09 rows=5216 width=8) (actual time=1.678..1.678 rows=5216 loops=1)
                    Buckets: 8192  Batches: 1  Memory Usage: 268kB
                    Buffers: shared hit=70
                    -> Seq Scan on orders o (cost=0.00..163.09 rows=5216 width=8) (actual time=0.004..1.034 rows=5216 loops=1)
                      Filter: (order_status = ANY (('{paid,shipped,delivered'}::text[]))
                      Rows Removed by Filter: 1554
                      Buffers: shared hit=70
                  -> Hash (cost=13.00..13.00 rows=600 width=19) (actual time=0.102..0.102 rows=600 loops=1)
                    Buckets: 1024  Batches: 1  Memory Usage: 39kB
                    Buffers: shared hit=7
                    -> Seq Scan on products p (cost=0.00..13.00 rows=600 width=19) (actual time=0.006..0.050 rows=600 loops=1)
                      Buffers: shared hit=7

Planning:
  Buffers: shared hit=26
Planning Time: 0.602 ms
Execution Time: 18.545 ms
(42개 행 )

skala_db=# \n
```

▲ 튜닝 후 — 인덱스/재작성 적용 후 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x39

skala_db=# \i q04_result.sql
SELECT product_id, product_name, revenue, RANK() OVER (ORDER BY revenue DESC) AS rnk
FROM (SELECT p.product_id, p.product_name, sum(oi.line_total) AS revenue
      FROM order_items oi JOIN orders o ON o.order_id=oi.order_id
      JOIN products p ON p.product_id=oi.product_id
      WHERE o.order_status IN ('paid','shipped','delivered')
      GROUP BY p.product_id, p.product_name) t
ORDER BY rnk LIMIT 20;

 product_id | product_name | revenue | rnk
-----
298 | Product 175 | 1628647.20 | 1
482 | Product 299 | 694075.65 | 2
215 | Product 574 | 16699.43 | 3
128 | Product 3 | 15689.36 | 4
602 | Product 199 | 15164.42 | 5
231 | Product 56 | 14842.64 | 6
564 | Product 218 | 13523.22 | 7
39 | Product 571 | 13366.56 | 8
125 | Product 493 | 12643.27 | 9
414 | Product 417 | 12439.87 | 10
41 | Product 41 | 12393.75 | 11
357 | Product 666 | 12328.56 | 12
380 | Product 267 | 12171.01 | 13
463 | Product 18 | 11074.59 | 14
197 | Product 334 | 11028.57 | 15
100 | Product 442 | 10969.47 | 16
131 | Product 153 | 10954.06 | 17
426 | Product 358 | 10722.15 | 18
261 | Product 425 | 10651.93 | 19
516 | Product 499 | 10575.90 | 20
(20개 행 )

skala_db=#
```

▲ 실행 결과 (튜닝 전후 동일 — 정확성 보존)

개선 — 필터가 상태뿐이라 order_items 전량 집계가 불가피하고 RANK() 정렬도 필수 → 전/후 계획 동일(계산 바운드). 근본 개선책은 사전 집계(제출물 3의 MV).

결과 — 1위 Product 175(162만), 2위 Product 299(69만) ... (전후 동일)

비즈니스 인사이트 — 상위 2개 제품(Product 175-299)이 162만·69만으로 압도적, 3위부터 급락 → 매출이 소수 히트상품에 편중(롱테일). 히트상품 품질 관리가 매출 방어 핵심.

Q5. RFM 분석 — 최근성/빈도/금액 쿼리 재작성

요구사항 — 고객별 recency_days, frequency(주문수), monetary(매출). 금액 상위 20. 선택 컬럼 4개.

병목 — frequency = count(DISTINCT order_id) → GroupAggregate 입력 (customer_id, order_id) 14,250행 Sort.

튜닝: 쿼리 재작성 — 주문 단위 선집계 후 고객별 재집계 (count(DISTINCT) 제거)

SQL — 튜닝 전 쿼리

```
SELECT o.customer_id,
       (now()::date - max(o.order_ts)::date) AS recency_days, -- R: 마지막 구매 후 경과일
       count(DISTINCT o.order_id) AS frequency, -- F: 주문 빈도(DISTINCT)
       sum(oi.line_total) AS monetary -- M: 누적 구매액
FROM   orders o
JOIN   order_items oi ON oi.order_id = o.order_id
WHERE  o.order_status IN ('paid','shipped','delivered')
GROUP BY o.customer_id
ORDER BY monetary DESC LIMIT 20; -- 금액 상위 20
```

SQL — 튜닝 후 (재작성 쿼리)

```
-- 주문 단위 선집계 후 고객별 재집계 (count(DISTINCT) 제거)
SELECT t.customer_id,
       (now()::date - max(t.order_ts)::date) AS recency_days, -- R: 최근성
       count(*) AS frequency, sum(t.order_rev) AS monetary -- F(=count*), M
FROM   (
  SELECT o.order_id, o.customer_id, o.order_ts, sum(oi.line_total) AS order_rev -- 주문별 선집계
  FROM   orders o JOIN order_items oi ON oi.order_id = o.order_id
  WHERE  o.order_status IN ('paid','shipped','delivered')
  GROUP BY o.order_id, o.customer_id, o.order_ts
) t
GROUP BY t.customer_id ORDER BY monetary DESC LIMIT 20; -- 고객별 재집계
```

```
skala_db=# \i q05_before.sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT o.customer_id, (now()::date - max(o.order_ts)::date) AS recency_days,
       count(DISTINCT o.order_id) AS frequency, sum(oi.line_total) AS monetary
FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
WHERE o.order_status IN ('paid','shipped','delivered')
GROUP BY o.customer_id ORDER BY monetary DESC LIMIT 20;

QUERY PLAN

Limit  (cost=1934.83..1934.88 rows=20 width=52) (actual time=14.191..14.196 rows=20 loops=1)
 Buffers: shared hit=248
-> Sort  (cost=1934.83..1942.06 rows=2892 width=52) (actual time=14.190..14.193 rows=20 loops=1)
   Sort Key: (sum(oi.line_total)) DESC
   Sort Method: top-N heapsort  Memory: 26kB
   Buffers: shared hit=248
-> GroupAggregate  (cost=1615.67..1857.88 rows=2892 width=52) (actual time=10.046..13.866 rows=2227 loops=1)
   Group Key: o.customer_id
   Buffers: shared hit=248
-> Sort  (cost=1615.67..1651.10 rows=14171 width=30) (actual time=9.996..10.536 rows=14250 loops=1)
   Sort Key: o.customer_id, o.order_id
   Sort Method: quicksort  Memory: 1052kB
   Buffers: shared hit=248
-> Hash Join  (cost=228.29..638.53 rows=14171 width=30) (actual time=2.182..7.075 rows=14250 loops=1)
   Hash Cond: (oi.order_id = o.order_id)
   Buffers: shared hit=248
-> Seq Scan on order_items oi  (cost=0.00..361.93 rows=18393 width=14) (actual time=0.007..1.270 rows=18393 loops=1)
   Buffers: shared hit=178
-> Hash  (cost=163.00..163.09 rows=5216 width=24) (actual time=2.142..2.143 rows=5216 loops=1)
   Buckets: 8192  Batches: 1  Memory Usage: 350kB
   Buffers: shared hit=70
-> Seq Scan on orders o  (cost=0.00..163.09 rows=5216 width=24) (actual time=0.008..1.612 rows=5216 loops=1)
   Filter: (order_status = ANY ('{paid,shipped,delivered}'))
   Rows Removed by Filter: 1554
   Buffers: shared hit=70

Planning:
 Buffers: shared hit=15
Planning Time: 0.343 ms
Execution Time: 14.248 ms
(29개 행)

skala_db=#
```

▲ 튜닝 전 — EXPLAIN (ANALYZE, BUFFERS) baseline 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x51

skala_db=# \i q05_after.sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT t.customer_id, (now()::date - max(t.order_ts)::date) AS recency_days,
       count(*) AS frequency, sum(t.order_rev) AS monetary
FROM (SELECT o.order_id, o.customer_id, o.order_ts, sum(oi.line_total) AS order_rev
      FROM orders o JOIN order_items oi ON oi.order_id=o.order_id
      WHERE o.order_status IN ('paid','shipped','delivered'))
GROUP BY o.order_id, o.customer_id, o.order_ts) t
ORDER BY t.customer_id ORDER BY monetary DESC LIMIT 20;

QUERY PLAN

Limit (cost=888.73..888.78 rows=20 width=52) (actual time=18.195..18.202 rows=20 loops=1)
  Buffers: shared hit=248
  -> Sort (cost=888.73..889.23 rows=200 width=52) (actual time=18.194..18.198 rows=20 loops=1)
    Sort Key: (sum((sum(oi.line_total)))) DESC
    Sort Method: top-N heapsort  Memory: 27kB
    Buffers: shared hit=248
    -> HashAggregate (cost=878.91..883.41 rows=200 width=52) (actual time=16.262..17.707 rows=2227 loops=1)
      Group Key: o.customer_id
      Batches: 1  Memory Usage: 1153kB
      Buffers: shared hit=248
      -> HashAggregate (cost=709.39..774.69 rows=5216 width=56) (actual time=12.118..14.289 rows=5216 loops=1)
        Group Key: o.order_id
        Batches: 1  Memory Usage: 2513kB
        Buffers: shared hit=248
        -> Hash Join (cost=228.29..638.53 rows=14171 width=30) (actual time=2.907..8.258 rows=14250 loops=1)
          Hash Cond: (oi.order_id = o.order_id)
          Buffers: shared hit=248
          -> Seq Scan on order_items oi (cost=0.00..361.93 rows=18393 width=14) (actual time=0.004..1.244 rows=18393 loops=1)
            Buffers: shared hit=178
          -> Hash (cost=163.09..163.09 rows=5216 width=24) (actual time=2.896..2.896 rows=5216 loops=1)
            Buckets: 8192  Batches: 1  Memory Usage: 350kB
            Buffers: shared hit=70
            -> Seq Scan on orders o (cost=0.00..163.09 rows=5216 width=24) (actual time=0.003..1.122 rows=5216 loops=1)
              Filter: (order_status = ANY ('{paid,shipped,delivered}'))
              Rows Removed by Filter: 1554
              Buffers: shared hit=70
Planning:
  Buffers: shared hit=14
Planning Time: 0.463 ms
Execution Time: 18.254 ms
(30개 행 )

skala_db=#
```

▲ 튜닝 후 — 인덱스/재작성 적용 후 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x39

skala_db=# \i q05_result.sql
SELECT t.customer_id, (now()::date - max(t.order_ts)::date) AS recency_days,
       count(*) AS frequency, sum(t.order_rev) AS monetary
FROM (SELECT o.order_id, o.customer_id, o.order_ts, sum(oi.line_total) AS order_rev
      FROM orders o JOIN order_items oi ON oi.order_id=o.order_id
      WHERE o.order_status IN ('paid','shipped','delivered'))
GROUP BY o.order_id, o.customer_id, o.order_ts) t
ORDER BY t.customer_id ORDER BY monetary DESC LIMIT 20;

customer_id | recency_days | frequency | monetary
-----
19          | 4            | 21        | 37420.49
8           | 7            | 21        | 38856.78
9           | 1            | 21        | 29539.44
22          | 2            | 21        | 28859.38
24          | 2            | 21        | 27989.41
18          | 3            | 21        | 26244.87
10          | 0            | 21        | 26213.05
21          | 8            | 21        | 25641.69
4           | 2            | 21        | 25159.75
15          | 2            | 21        | 25131.91
26          | 1            | 21        | 23983.57
25          | 0            | 21        | 23852.45
12          | 2            | 21        | 23437.73
20          | 0            | 21        | 22897.52
2           | 1            | 21        | 22735.72
7           | 0            | 21        | 22298.63
6           | 1            | 21        | 22133.47
1           | 3            | 21        | 21219.52
3           | 2            | 21        | 21837.04
17          | 1            | 21        | 20941.21

(20개 행 )

skala_db=#
```

▲ 실행 결과 (튜닝 전후 동일 — 정확성 보존)

- 개선 — GroupAggregate + Sort(14,250행) → 2단 HashAggregate. count(DISTINCT) 제거로 대형 정렬 노드 제거.
- 결과 — 헤비고객 19번(37,420) ~ 17번(20,941), frequency 21 균일 (전후 동일)
- 비즈니스 인사이트 — 상위 20 VIP가 frequency 21로 균일하게 높고 monetary 2만~3.7만 → 헤비 고객층이 뚜렷. 이들 대상 리텐션·VIP 혜택 설계가 효과적.

Q6. 첫 구매 후 30일 내 재구매율 인덱스 추가

요구사항 — 고객 첫 결제완료 이후 30일 이내 재주문 비율(%). 선택 컬럼: cohort, repurchased, pct_30d.

병목 — LATERAL 프로브가 idx_orders_customer_ts로 Index Scan하나 status 필터를 heap에서 재확인(버퍼 hit 5,539).

튜닝: 복합 부분 인덱스 — status 조건까지 인덱스에 포함

SQL — 튜닝 전 쿼리

```
WITH firsts AS (  
    SELECT customer_id, min(order_ts) AS first_ts          -- 고객별 첫 구매 시점 산출  
    FROM orders WHERE order_status IN ('paid','shipped','delivered')  
    GROUP BY customer_id  
)  
SELECT count(*) AS cohort,                                -- 첫 구매 고객 수(모수)  
       count(*) FILTER (WHERE re.customer_id IS NOT NULL) AS repurchased, -- 30일내 재구매자  
       round(100.0*count(*) FILTER (WHERE re.customer_id IS NOT NULL)/count(*),2) AS pct_30d -- 재구매율%  
FROM   firsts f  
LEFT JOIN LATERAL (  
    SELECT o2.customer_id FROM orders o2                  -- 각 첫구매행마다 재주문 탐색(프로브)  
    WHERE o2.customer_id = f.customer_id                  -- 동일 고객  
      AND o2.order_status IN ('paid','shipped','delivered')  
      AND o2.order_ts > f.first_ts                        -- 첫 구매 이후  
      AND o2.order_ts <= f.first_ts + interval '30 days' -- 30일 이내  
    LIMIT 1                                                -- 1건이면 재구매 성립 → 조기 종료  
) re ON true;
```

SQL — 튜닝 후 (인덱스 DDL)

```
-- status 조건까지 담은 복합 부분 인덱스, 쿼리는 동일 (Index Only Scan 유도)  
CREATE INDEX idx_orders_cust_rev_ts ON orders(customer_id, order_ts) -- 프로브 복합키  
WHERE order_status IN ('paid','shipped','delivered');              -- 매출 상태만
```

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x44  
skala_db=# \i q06.before.sql  
EXPLAIN (ANALYZE, BUFFERS)  
WITH firsts AS (SELECT customer_id, min(order_ts) AS first_ts FROM orders  
WHERE order_status IN ('paid','shipped','delivered') GROUP BY customer_id)  
SELECT count(*) AS cohort,  
       count(*) FILTER (WHERE re.customer_id IS NOT NULL) AS repurchased,  
       round(100.0*count(*) FILTER (WHERE re.customer_id IS NOT NULL)/count(*),2) AS pct_30d  
FROM firsts f LEFT JOIN LATERAL (  
    SELECT o2.customer_id FROM orders o2  
    WHERE o2.customer_id=f.customer_id AND o2.order_status IN ('paid','shipped','delivered')  
      AND o2.order_ts>f.first_ts AND o2.order_ts<=f.first_ts+interval '30 days' LIMIT 1) re ON true;  
QUERY PLAN  
-----  
Aggregate (cost=24297.60..24297.62 rows=1 width=48) (actual time=5.996..5.997 rows=1 loops=1)  
  Buffers: shared hit=5609  
  -> Nested Loop Left Join (cost=189.45..24283.14 rows=2892 width=8) (actual time=2.172..5.860 rows=2227 loops=1)  
    Buffers: shared hit=5609  
    -> HashAggregate (cost=189.17..218.09 rows=2892 width=16) (actual time=2.127..2.449 rows=2227 loops=1)  
      Group Key: orders.customer_id  
      Batches: 1 Memory Usage: 369kB  
      Buffers: shared hit=70  
    -> Seq Scan on orders (cost=0.00..163.09 rows=5216 width=16) (actual time=0.006..0.816 rows=5216 loops=1)  
      Filter: (order_status = ANY ('{paid,shipped,delivered}'))  
      Rows Removed by Filter: 1554  
      Buffers: shared hit=70  
    -> Limit (cost=0.29..8.31 rows=1 width=8) (actual time=0.001..0.001 rows=0 loops=2227)  
      Buffers: shared hit=5539  
      -> Index Scan using idx_orders_customer_ts on orders o2 (cost=0.29..8.31 rows=1 width=8) (actual time=0.001..0.001 rows=0 loops=2227)  
        Index Cond: ((customer_id = orders.customer_id) AND (order_ts > (min(orders.order_ts))) AND (order_ts <= ((min(orders.order_ts)) + '30 days')))  
        Filter: (order_status = ANY ('{paid,shipped,delivered}'))  
        Buffers: shared hit=5539  
Planning:  
  Buffers: shared hit=12  
  Planning Time: 0.184 ms  
  Execution Time: 6.033 ms  
(22개 행)  
skala_db=#
```

▲ 튜닝 전 — EXPLAIN (ANALYZE, BUFFERS) baseline 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x44

skala_db=# \i q06_after.sql
EXPLAIN (ANALYZE, BUFFERS)
WITH firsts AS (SELECT customer_id, min(order_ts) AS first_ts FROM orders
WHERE order_status IN ('paid','shipped','delivered') GROUP BY customer_id)
SELECT count(*) AS cohort,
count(*) FILTER (WHERE re.customer_id IS NOT NULL) AS repurchased,
round(100.0*count(*) FILTER (WHERE re.customer_id IS NOT NULL)/count(*),2) AS pct_30d
FROM firsts f LEFT JOIN LATERAL (
SELECT o2.customer_id FROM orders o2
WHERE o2.customer_id=f.customer_id AND o2.order_status IN ('paid','shipped','delivered')
AND o2.order_ts>f.first_ts AND o2.order_ts<=f.first_ts+interval '30 days' LIMIT 1) re ON true;
QUERY PLAN

-----
Aggregate  (cost=12718.76..12718.78 rows=1 width=48) (actual time=9.927..9.928 rows=1 loops=1)
  Buffers: shared hit=4530
  -> Nested Loop Left Join  (cost=189.45..12704.30 rows=2892 width=8) (actual time=4.353..9.684 rows=2227 loops=1)
    Buffers: shared hit=4530
    -> HashAggregate  (cost=189.17..218.09 rows=2892 width=16) (actual time=4.315..4.770 rows=2227 loops=1)
      Group Key: orders.customer_id
      Batches: 1  Memory Usage: 369kB
      Buffers: shared hit=70
      -> Seq Scan on orders  (cost=0.00..163.09 rows=5216 width=16) (actual time=0.010..2.724 rows=5216 loops=1)
        Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
        Rows Removed by Filter: 1554
        Buffers: shared hit=70
    -> Limit  (cost=0.29..4.31 rows=1 width=8) (actual time=0.002..0.002 rows=0 loops=2227)
      Buffers: shared hit=4460
      -> Index Only Scan using idx_orders_cust_rev_ts on orders o2  (cost=0.29..4.31 rows=1 width=8) (actual time=0.001..0.001 rows=0 loops=2227)
        Index Cond: ((customer_id = orders.customer_id) AND (order_ts > (min(orders.order_ts))) AND (order_ts <= ((min(orders.order_ts)) + '30 days'::interval)))
        Heap Fetches: 0
        Buffers: shared hit=2460
Planning Time: 0.201 ms
Execution Time: 9.977 ms
(20개 행 )

skala_db=#
```

▲ 튜닝 후 — 인덱스/재작성 적용 후 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x20

skala_db=# \i q06_result.sql
WITH firsts AS (SELECT customer_id, min(order_ts) AS first_ts FROM orders
WHERE order_status IN ('paid','shipped','delivered') GROUP BY customer_id)
SELECT count(*) AS cohort,
count(*) FILTER (WHERE re.customer_id IS NOT NULL) AS repurchased,
round(100.0*count(*) FILTER (WHERE re.customer_id IS NOT NULL)/count(*),2) AS pct_30d
FROM firsts f LEFT JOIN LATERAL (
SELECT o2.customer_id FROM orders o2
WHERE o2.customer_id=f.customer_id AND o2.order_status IN ('paid','shipped','delivered')
AND o2.order_ts>f.first_ts AND o2.order_ts<=f.first_ts+interval '30 days' LIMIT 1) re ON true;
cohort | repurchased | pct_30d
-----+-----+-----
2227 | 1085 | 48.72
(1개 행 )

skala_db=#
```

▲ 실행 결과 (튜닝 전후 동일 — 정확성 보존)

개선 — 재구매 탐색 프로브가 Index Scan(heap 재확인, Buffers hit 5,539) → Index Only Scan (idx_orders_cust_rev_ts)로 전환 → heap 접근 제거, 버퍼 대폭 감소.

결과 — cohort 2,227 / repurchased 1,085 / pct_30d 48.72% (전후 동일)

비즈니스 인사이트 — 첫 구매 후 30일 내 재구매율 48.72% → 신규의 절반이 한 달 안에 재구매. 초기 리텐션이 양호하므로 첫 구매 직후 30일 온보딩·리마케팅에 집중할 가치가 큼.

Q7. 재고 임계치 미만 상품 (곧 품질 위험) 인덱스 추가

요구사항 — qty_on_hand < reorder_point 상품, 부족량 큰 순. 선택 컬럼: product_id, name, qty, reorder_point, shortage.

병목 — inventory 전량(600) Seq Scan 후 두 컬럼 비교 필터(600→61).

튜닝: 부분 인덱스 — 두 컬럼 비교 조건을 그대로 인덱스 조건으로 (품질 위험 상품만 인덱싱)

SQL — 튜닝 전 쿼리

```
SELECT i.product_id, p.product_name, i.qty_on_hand, i.reorder_point,
       (i.reorder_point - i.qty_on_hand) AS shortage    -- 부족량 = 재주문점 - 현재고
FROM   inventory i
JOIN   products p ON p.product_id = i.product_id      -- 상품명 조인
WHERE  i.qty_on_hand < i.reorder_point                -- 품질 위험 (현재고 < 재주문점)
ORDER BY shortage DESC LIMIT 10;                     -- 부족량 큰 순 (상위 10개 표시, 전체 위험 61건)
```

SQL — 튜닝 후 (인덱스 DDL)

```
-- 두 컬럼 비교 조건을 그대로 담은 부분 인덱스, 쿼리는 동일
CREATE INDEX idx_inventory_low_stock ON inventory(product_id) -- 위험 상품만 인덱싱
WHERE qty_on_hand < reorder_point;
-- (600행 소형이라 옵티마이저는 Seq Scan 선택; 인덱스 활용 계획은
-- SET enable_seqscan=off로 검증 → Bitmap Index Scan 61행)
```

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x39

skala_db=# \i q07.before.sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT i.product_id, p.product_name, i.qty_on_hand, i.reorder_point,
       (i.reorder_point-i.qty_on_hand) AS shortage
FROM inventory i JOIN products p ON p.product_id=i.product_id
WHERE i.qty_on_hand < i.reorder_point ORDER BY shortage DESC LIMIT 10;
                                QUERY PLAN

Limit  (cost=34.41..34.43 rows=10 width=31) (actual time=0.182..0.184 rows=10 loops=1)
  Buffers: shared hit=15
  -> Sort  (cost=34.41..34.91 rows=200 width=31) (actual time=0.182..0.183 rows=10 loops=1)
        Sort Key: ((i.reorder_point - i.qty_on_hand)) DESC
        Sort Method: top-N heapsort  Memory: 26kB
        Buffers: shared hit=15
  -> Hash Join  (cost=15.00..30.09 rows=200 width=31) (actual time=0.059..0.161 rows=61 loops=1)
        Hash Cond: (p.product_id = i.product_id)
        Buffers: shared hit=12
        -> Seq Scan on products p  (cost=0.00..13.00 rows=600 width=19) (actual time=0.005..0.049 rows=600 loops=1)
              Buffers: shared hit=7
        -> Hash  (cost=12.50..12.50 rows=200 width=16) (actual time=0.049..0.049 rows=61 loops=1)
              Buckets: 1024  Batches: 1  Memory Usage: 11kB
              Buffers: shared hit=5
        -> Seq Scan on inventory i  (cost=0.00..12.50 rows=200 width=16) (actual time=0.004..0.038 rows=61 loops=1)
              Filter: (qty_on_hand < reorder_point)
              Rows Removed by Filter: 539
              Buffers: shared hit=5

Planning:
  Buffers: shared hit=50
Planning Time: 0.400 ms
Execution Time: 0.206 ms
(22개 행)

skala_db=#
```

▲ 튜닝 전 — EXPLAIN (ANALYZE, BUFFERS) baseline 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x46

skala_db=# \i q07_after.sql
SET enable_seqscan = off;
SET
EXPLAIN (ANALYZE, BUFFERS)
SELECT i.product_id, p.product_name, i.qty_on_hand, i.reorder_point,
       (i.reorder_point-i.qty_on_hand) AS shortage
FROM inventory i JOIN products p ON p.product_id=i.product_id
WHERE i.qty_on_hand < i.reorder_point ORDER BY shortage DESC LIMIT 10;

QUERY PLAN

Limit (cost=60.18..60.20 rows=10 width=31) (actual time=0.150..0.152 rows=10 loops=1)
  Buffers: shared hit=16
  -> Sort (cost=60.18..60.68 rows=200 width=31) (actual time=0.149..0.150 rows=10 loops=1)
    Sort Key: ((i.reorder_point - i.qty_on_hand)) DESC
    Sort Method: top-N heapsort  Memory: 26kB
    Buffers: shared hit=16
    -> Hash Join (cost=18.77..55.86 rows=200 width=31) (actual time=0.057..0.139 rows=61 loops=1)
      Hash Cond: (p.product_id = i.product_id)
      Buffers: shared hit=16
      -> Index Scan using products_pkey on products p (cost=0.28..35.27 rows=600 width=19) (actual time=0.006..0.053 rows=600 loops=1)
        Buffers: shared hit=10
      -> Hash (cost=16.00..16.00 rows=200 width=16) (actual time=0.043..0.044 rows=61 loops=1)
        Buckets: 1024  Batches: 1  Memory Usage: 11kB
        Buffers: shared hit=6
        -> Bitmap Heap Scan on inventory i (cost=8.50..16.00 rows=200 width=16) (actual time=0.018..0.031 rows=61 loops=1)
          Recheck Cond: (qty_on_hand < reorder_point)
          Heap Blocks: exact=5
          Buffers: shared hit=6
          -> Bitmap Index Scan on idx_inventory_low_stock (cost=0.00..8.45 rows=200 width=0) (actual time=0.012..0.012 rows=61 loops=1)
            Buffers: shared hit=1

Planning:
  Buffers: shared hit=40
Planning Time: 0.395 ms
Execution Time: 0.168 ms
(247개 행)

RESET enable_seqscan;
RESET
skala_db=#
```

▲ 튜닝 후 — 인덱스/재작성 적용 후 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x25

skala_db=# \i q07_result.sql
SELECT i.product_id, p.product_name, i.qty_on_hand, i.reorder_point,
       (i.reorder_point-i.qty_on_hand) AS shortage
FROM inventory i JOIN products p ON p.product_id=i.product_id
WHERE i.qty_on_hand < i.reorder_point ORDER BY shortage DESC LIMIT 10;

 product_id | product_name | qty_on_hand | reorder_point | shortage
-----
 586 | Product 459 |          0 |          30 |        30
 285 | Product 435 |          0 |          30 |        30
 56 | Product 141 |          0 |          30 |        30
 655 | Product 468 |          0 |          30 |        30
 257 | Product 65 |          0 |          30 |        30
 39 | Product 871 |          1 |          30 |        29
 99 | Product 192 |          1 |          30 |        29
 98 | Product 452 |          1 |          30 |        29
 14 | Product 81 |          1 |          30 |        29
 262 | Product 95 |          1 |          30 |        29
(10개 행)

skala_db=#
```

▲ 실행 결과 (튜닝 전후 동일 — 정확성 보존)

- 개선 — Seq Scan(600행 전수) → Bitmap Index Scan (idx_inventory_low_stock, 61행). 단, 600행 소형이라 두 방식의 절대시간 차이는 측정 노이즈 수준(sub-ms)이며, 옵티마이저는 비용 추정이 더 낮은 Seq Scan을 선택 — 접근 행수↓ 이득은 재고가 커질수록 뚜렷해진다.
- 결과 — 품질 위험 61건, 상위는 재고 0 / reorder_point 30 (shortage 30)
- 비즈니스 인사이트 — 품질 위험 61건, 그중 재고 0이 5건 → 즉시 발주 대상. 부족량(shortage) 내림차순으로 발주 우선순위를 바로 산출.

Q8. 효자상품 — 평점 4.5↑ & 리뷰 50↑ 인덱스 추가

요구사항 — HAVING avg(rating)>=4.5 AND count(*)>=50. 선택 컬럼: product_id, name, avg_rating, review_cnt.

병목 — reviews 전량(2,064) Seq Scan → products 조인 → HashAggregate + HAVING.

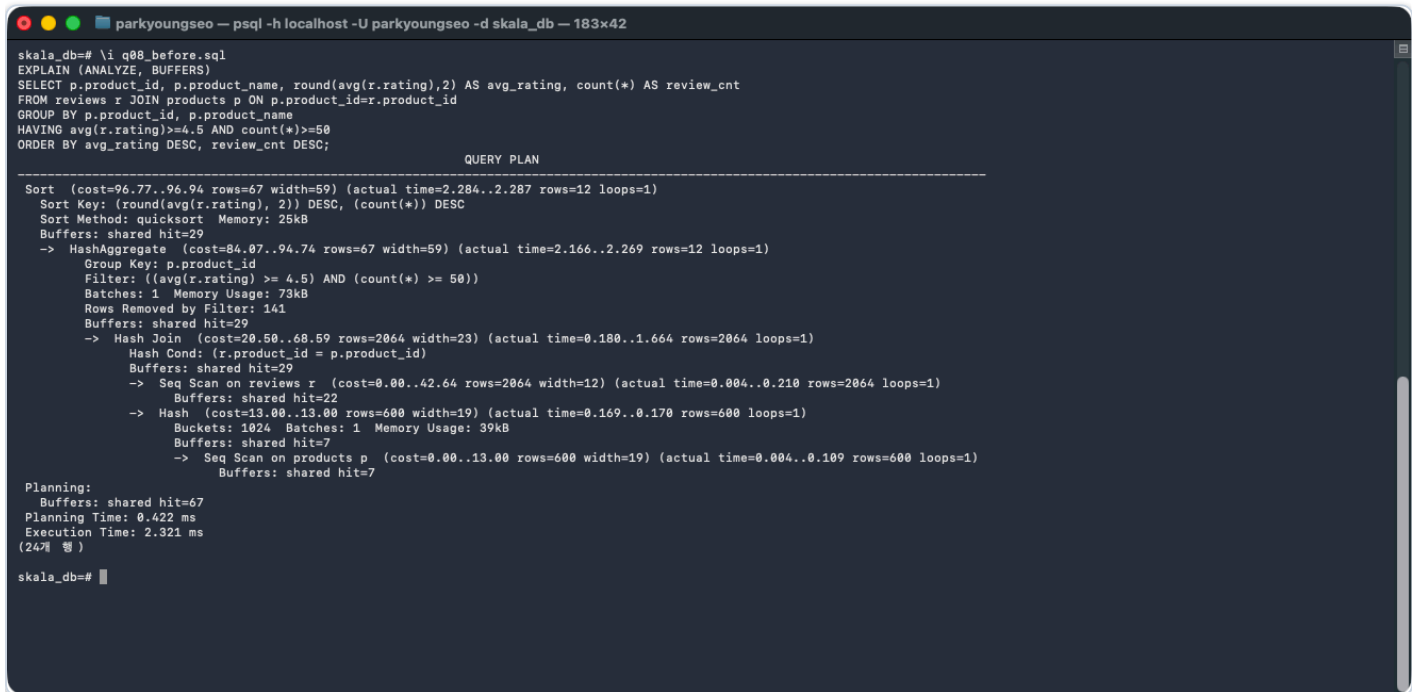
튜닝: 커버링 인덱스 — product_id로 그룹, rating은 INCLUDE (Index Only Scan)

SQL — 튜닝 전 쿼리

```
SELECT p.product_id, p.product_name,
       round(avg(r.rating),2) AS avg_rating, count(*) AS review_cnt  -- 평균평점·리뷰수
FROM   reviews r
JOIN   products p ON p.product_id = r.product_id
GROUP BY p.product_id, p.product_name
HAVING avg(r.rating) >= 4.5 AND count(*) >= 50  -- 효자상품: 고평점 & 표본 충분
ORDER BY avg_rating DESC, review_cnt DESC;
```

SQL — 튜닝 후 (인덱스 DDL)

```
-- product_id로 그룹, rating은 INCLUDE 한 커버링 인덱스, 쿼리는 동일
CREATE INDEX idx_reviews_prod_rating ON reviews(product_id) INCLUDE (rating);
-- (Index Only Scan 계획은 SET enable_seqscan=off로 검증 → Heap Fetches 0)
```



```
scala_db=# \i q08_before.sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT p.product_id, p.product_name, round(avg(r.rating),2) AS avg_rating, count(*) AS review_cnt
FROM reviews r JOIN products p ON p.product_id=r.product_id
GROUP BY p.product_id, p.product_name
HAVING avg(r.rating)>=4.5 AND count(*)>=50
ORDER BY avg_rating DESC, review_cnt DESC;

               QUERY PLAN

Sort  (cost=96.77..96.94 rows=67 width=59) (actual time=2.284..2.287 rows=12 loops=1)
  Sort Key: (round(avg(r.rating), 2)) DESC, (count(*)) DESC
  Sort Method: quicksort  Memory: 25kB
  Buffers: shared hit=29
   -> HashAggregate  (cost=84.07..94.74 rows=67 width=59) (actual time=2.166..2.269 rows=12 loops=1)
     Group Key: p.product_id
     Filter: ((avg(r.rating) >= 4.5) AND (count(*) >= 50))
     Batches: 1  Memory Usage: 73kB
     Rows Removed by Filter: 141
     Buffers: shared hit=29
      -> Hash Join  (cost=20.50..60.59 rows=2064 width=23) (actual time=0.180..1.664 rows=2064 loops=1)
        Hash Cond: (r.product_id = p.product_id)
        Buffers: shared hit=29
         -> Seq Scan on reviews r  (cost=0.00..42.64 rows=2064 width=12) (actual time=0.004..0.210 rows=2064 loops=1)
            Buffers: shared hit=22
         -> Hash  (cost=13.00..13.00 rows=600 width=19) (actual time=0.169..0.170 rows=600 loops=1)
            Buckets: 1024  Batches: 1  Memory Usage: 39kB
            Buffers: shared hit=7
             -> Seq Scan on products p  (cost=0.00..13.00 rows=600 width=19) (actual time=0.004..0.109 rows=600 loops=1)
                Buffers: shared hit=7

Planning:
  Buffers: shared hit=67
  Planning Time: 0.422 ms
  Execution Time: 2.321 ms
(24개 행)

scala_db=#
```

▲ 튜닝 전 — EXPLAIN (ANALYZE, BUFFERS) baseline 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x48
skala_db=# \i q08_after.sql
SET enable_seqscan = off;
SET
EXPLAIN (ANALYZE, BUFFERS)
SELECT p.product_id, p.product_name, round(avg(r.rating),2) AS avg_rating, count(*) AS review_cnt
FROM reviews r JOIN products p ON p.product_id=r.product_id
GROUP BY p.product_id, p.product_name
HAVING avg(r.rating)>=4.5 AND count(*)>=50
ORDER BY avg_rating DESC, review_cnt DESC;

QUERY PLAN

Sort (cost=147.65..147.81 rows=67 width=59) (actual time=1.502..1.504 rows=12 loops=1)
  Sort Key: (round(avg(r.rating), 2)) DESC, (count(*)) DESC
  Sort Method: quicksort Memory: 25kB
  Buffers: shared hit=20
  -> HashAggregate (cost=134.95..145.61 rows=67 width=59) (actual time=1.451..1.493 rows=12 loops=1)
    Group Key: p.product_id
    Filter: ((avg(r.rating) >= 4.5) AND (count(*) >= 50))
    Batches: 1 Memory Usage: 73kB
    Rows Removed by Filter: 141
    Buffers: shared hit=20
    -> Hash Join (cost=43.05..119.47 rows=2064 width=23) (actual time=0.213..1.146 rows=2064 loops=1)
      Hash Cond: (r.product_id = p.product_id)
      Buffers: shared hit=20
      -> Index Only Scan using idx_reviews_prod_rating on reviews r (cost=0.28..71.24 rows=2064 width=12) (actual time=0.005..0.624 rows=2064 loops=1)
        Heap Fetches: 0
        Buffers: shared hit=10
      -> Hash (cost=35.27..35.27 rows=600 width=19) (actual time=0.205..0.205 rows=600 loops=1)
        Buckets: 1024 Batches: 1 Memory Usage: 39kB
        Buffers: shared hit=10
        -> Index Scan using products_pkey on products p (cost=0.28..35.27 rows=600 width=19) (actual time=0.075..0.147 rows=600 loops=1)
          Buffers: shared hit=10

Planning:
  Buffers: shared hit=42
Planning Time: 0.349 ms
Execution Time: 1.529 ms
(25개 행 )

RESET enable_seqscan;
RESET
skala_db=#
```

▲ 튜닝 후 — 인덱스/재작성 적용 후 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x28
skala_db=# \i q08_result.sql
SELECT p.product_id, p.product_name, round(avg(r.rating),2) AS avg_rating, count(*) AS review_cnt
FROM reviews r JOIN products p ON p.product_id=r.product_id
GROUP BY p.product_id, p.product_name
HAVING avg(r.rating)>=4.5 AND count(*)>=50
ORDER BY avg_rating DESC, review_cnt DESC;
 product_id | product_name | avg_rating | review_cnt
-----
 259 | Product 265 |      4.83 |        160
 293 | Product 187 |      4.83 |        160
 690 | Product 538 |      4.81 |        160
 579 | Product 898 |      4.81 |        160
 379 | Product 327 |      4.78 |        161
 41 | Product 41 |      4.78 |        160
 45 | Product 371 |      4.77 |        160
 181 | Product 264 |      4.77 |        160
 136 | Product 163 |      4.76 |        161
 444 | Product 28 |      4.76 |        160
 358 | Product 496 |      4.73 |        161
 157 | Product 233 |      4.71 |        160
(12개 행 )
skala_db=#
```

▲ 실행 결과 (튜닝 전후 동일 — 정확성 보존)

- 개선 — Seq Scan → Index Only Scan (Heap Fetches: 0) — (product_id, rating)만 읽어 heap I/O 제거. 2,064행 규모라 절대 시간은 유사하나 리뷰 폭증 대비 확장성 확보.
- 결과 — 효자상품 12개 (avg 4.71~4.83, 리뷰 160~161개, 전후 동일)
- 비즈니스 인사이트 — 효자상품 12개(평점 4.71~4.83, 리뷰 160+) → 검증된 고평점·고리뷰 상품. 메인 노출·번들·리뷰 마케팅의 1순위 후보.

Q9. 쿠폰 사용 영향 — 쿠폰 有/無 AOV 비교 쿼리 재작성

요구사항 — coupon_code 유무별 주문수·AOV. 선택 컬럼: used_coupon, orders, aov.

병목 — 주문 단위 집계 후 바깥 GroupAggregate가 (used_coupon, order_id) 기준 Sort(5,216행) → 불필요 정렬.

튜닝: 쿼리 재작성 — 안쪽에서 (order_id, used_coupon) 선집계 → 바깥 HashAggregate

SQL — 튜닝 전 쿼리

```
SELECT (o.coupon_code IS NOT NULL) AS used_coupon,      -- 쿠폰 사용 여부(T/F)
       count(DISTINCT o.order_id) AS orders,           -- 주문수(DISTINCT)
       round(avg(order_rev),2) AS aov                 -- 평균 주문금액
FROM (
  SELECT o.order_id, o.coupon_code, sum(oi.line_total) AS order_rev -- 주문 단위 선집계
  FROM   orders o JOIN order_items oi ON oi.order_id = o.order_id
  WHERE  o.order_status IN ('paid','shipped','delivered')
  GROUP BY o.order_id, o.coupon_code
) o
GROUP BY 1;                                           -- 쿠폰 유무로 그룹
```

SQL — 튜닝 후 (재작성 쿼리)

```
-- 안쪽에서 (order_id, used_coupon) 선집계 → 바깥 HashAggregate (외부 Sort 제거)
SELECT used_coupon, count(*) AS orders, round(avg(order_rev),2) AS aov -- 쿠폰유무별 집계
FROM (
  SELECT o.order_id, (o.coupon_code IS NOT NULL) AS used_coupon,
         sum(oi.line_total) AS order_rev -- 주문 단위 선집계
  FROM   orders o JOIN order_items oi ON oi.order_id = o.order_id
  WHERE  o.order_status IN ('paid','shipped','delivered')
  GROUP BY o.order_id, (o.coupon_code IS NOT NULL)
) t
GROUP BY used_coupon;
```

```
parkyoungseo ~$ psql -h localhost -U parkyoungseo -d skala_db - 183x50
skala_db=# \i q09_before.sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT (o.coupon_code IS NOT NULL) AS used_coupon,
       count(DISTINCT o.order_id) AS orders, round(avg(order_rev),2) AS aov
FROM (SELECT o.order_id, o.coupon_code, sum(oi.line_total) AS order_rev
      FROM orders o JOIN order_items oi ON oi.order_id=o.order_id
      WHERE o.order_status IN ('paid','shipped','delivered')
      GROUP BY o.order_id, o.coupon_code) o
GROUP BY 1;

QUERY PLAN

GroupAggregate (cost=1148.80..1280.99 rows=2 width=41) (actual time=21.532..21.763 rows=2 loops=1)
  Group Key: ((o.coupon_code IS NOT NULL))
  Buffers: shared hit=253
  -> Sort (cost=1148.80..1161.84 rows=5216 width=41) (actual time=20.810..21.014 rows=5216 loops=1)
    Sort Key: ((o.coupon_code IS NOT NULL)), o.order_id
    Sort Method: quicksort Memory: 396kB
    Buffers: shared hit=253
    -> Subquery Scan on o (cost=709.39..826.75 rows=5216 width=41) (actual time=15.734..17.838 rows=5216 loops=1)
      Buffers: shared hit=248
      -> HashAggregate (cost=709.39..774.59 rows=5216 width=47) (actual time=15.733..17.356 rows=5216 loops=1)
        Group Key: o_1.order_id
        Batches: 1 Memory Usage: 2257kB
        Buffers: shared hit=248
        -> Hash Join (cost=228.29..638.53 rows=14171 width=21) (actual time=3.758..10.783 rows=14250 loops=1)
          Hash Cond: (oi.order_id = o_1.order_id)
          Buffers: shared hit=248
          -> Seq Scan on order_items oi (cost=0.00..361.93 rows=18393 width=14) (actual time=0.004..1.714 rows=18393 loops=1)
            Buffers: shared hit=178
          -> Hash (cost=163.09..163.09 rows=5216 width=15) (actual time=3.744..3.744 rows=5216 loops=1)
            Buckets: 8192 Batches: 1 Memory Usage: 277kB
            Buffers: shared hit=70
            -> Seq Scan on orders o_1 (cost=0.00..163.09 rows=5216 width=15) (actual time=0.004..2.711 rows=5216 loops=1)
              Filter: (order_status = ANY ('{paid,shipped,delivered}':text[]))
              Rows Removed by Filter: 1554
              Buffers: shared hit=70
        Planning:
          Buffers: shared hit=24
          Planning Time: 0.191 ms
          Execution Time: 21.810 ms
          (29% 행)

skala_db=#
```

▲ 튜닝 전 — EXPLAIN (ANALYZE, BUFFERS) baseline 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x43

skala_db=# \i q09_after.sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT used_coupon, count(*) AS orders, round(avg(order_rev),2) AS aov
FROM (SELECT o.order_id, (o.coupon_code IS NOT NULL) AS used_coupon, sum(oi.line_total) AS order_rev
FROM orders o JOIN order_items oi ON oi.order_id=o.order_id
WHERE o.order_status IN ('paid','shipped','delivered')
GROUP BY o.order_id, (o.coupon_code IS NOT NULL)) t
GROUP BY used_coupon;

QUERY PLAN

HashAggregate (cost=1057.78..1857.81 rows=2 width=41) (actual time=17.957..17.959 rows=2 loops=1)
  Group Key: ((o.coupon_code IS NOT NULL))
  Batches: 1 Memory Usage: 24kB
  Buffers: shared hit=248
  -> HashAggregate (cost=744.82..875.22 rows=10432 width=41) (actual time=15.487..17.184 rows=5216 loops=1)
    Group Key: o.order_id, (o.coupon_code IS NOT NULL)
    Batches: 1 Memory Usage: 2449kB
    Buffers: shared hit=248
    -> Hash Join (cost=228.29..638.53 rows=14171 width=15) (actual time=5.107..10.997 rows=14250 loops=1)
      Hash Cond: (oi.order_id = o.order_id)
      Buffers: shared hit=248
      -> Seq Scan on order_items oi (cost=0.00..361.93 rows=18393 width=14) (actual time=0.006..1.523 rows=18393 loops=1)
        Buffers: shared hit=178
      -> Hash (cost=163.09..163.09 rows=5216 width=15) (actual time=5.089..5.090 rows=5216 loops=1)
        Buckets: 8192 Batches: 1 Memory Usage: 277kB
        Buffers: shared hit=70
        -> Seq Scan on orders o (cost=0.00..163.09 rows=5216 width=15) (actual time=0.007..3.993 rows=5216 loops=1)
          Filter: (order_status = ANY ('{paid,shipped,delivered}'))
          Rows Removed by Filter: 1554
          Buffers: shared hit=70
Planning:
  Buffers: shared hit=15
Planning Time: 0.324 ms
Execution Time: 18.003 ms
(24개 행)

skala_db=#
```

▲ 튜닝 후 — 인덱스/재작성 적용 후 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x18

skala_db=# \i q09_result.sql
SELECT used_coupon, count(*) AS orders, round(avg(order_rev),2) AS aov
FROM (SELECT o.order_id, (o.coupon_code IS NOT NULL) AS used_coupon, sum(oi.line_total) AS order_rev
FROM orders o JOIN order_items oi ON oi.order_id=o.order_id
WHERE o.order_status IN ('paid','shipped','delivered')
GROUP BY o.order_id, (o.coupon_code IS NOT NULL)) t
GROUP BY used_coupon ORDER BY used_coupon,
      aov;

used_coupon | orders | aov
-----
f           | 3953   | 635.61
t           | 1263   | 1827.10
(2개 행)

skala_db=#
```

▲ 실행 결과 (튜닝 전후 동일 — 정확성 보존)

- 개선 — GroupAggregate + Sort(5,216행) → HashAggregate. 불필요한 외부 정렬 노드 제거.
- 결과 — 쿠폰無 3,953건 AOV 635.61 vs 쿠폰有 1,263건 AOV 1,827.10 (SAVE10이 고액주문에 집중)
- 비즈니스 인사이트 — 쿠폰 사용 주문 AOV 1,827 vs 미사용 635 — 약 2.9배. 쿠폰이 고액 주문에 집중되어 업셀링·객단가 상승 효과를 시사(쿠폰=단순 할인이 아니라 큰 장바구니를 유도).

Q10. 상위 1% 고객의 최근 60일 매출 인덱스 추가

요구사항 — 최근 60일 매출 상위 1%(99퍼센타일↑) 고객. 선택 컬럼: customer_id, rev.

병목 — 최근 60일 매출을 고객별 집계. orders를 (상태 AND 60일) 필터로 Seq Scan(3,894행 제거).

튜닝: 부분 인덱스(idx_orders_rev_ts) 활용 — 60일=약 42%로 Q3보다 선택적

SQL — 튜닝 전 쿼리

```
WITH cust_rev AS (                                -- 최근 60일 고객별 매출
  SELECT o.customer_id, sum(oi.line_total) AS rev
  FROM   orders o JOIN order_items oi ON oi.order_id = o.order_id
  WHERE  o.order_status IN ('paid','shipped','delivered')
        AND o.order_ts >= now() - interval '60 days' -- 최근 60일
  GROUP BY o.customer_id
)
SELECT customer_id, rev FROM cust_rev
WHERE rev >= (SELECT percentile_cont(0.99) WITHIN GROUP (ORDER BY rev) FROM cust_rev) -- 상위1% 임계값 이상
ORDER BY rev DESC;
```

SQL — 튜닝 후 (인덱스 DDL)

```
-- Q1과 동일 부분 인덱스 재사용(60일 ~ 42%로 선택적), 쿼리는 동일
CREATE INDEX idx_orders_rev_ts ON orders(order_ts) -- order_ts 정렬키
WHERE order_status IN ('paid','shipped','delivered'); -- 매출 상태만(부분 인덱스)
```

```
parkyoungseo - psql -h localhost -U parkyoungseo -d skala_db - 183x54
skala_db=# \i q10_before.sql
EXPLAIN (ANALYZE, BUFFERS)
WITH cust_rev AS (SELECT o.customer_id, sum(oi.line_total) AS rev
  FROM orders o JOIN order_items oi ON oi.order_id=o.order_id
  WHERE o.order_status IN ('paid','shipped','delivered')
        AND o.order_ts >= now() - interval '60 days' GROUP BY o.customer_id)
SELECT customer_id, rev FROM cust_rev
WHERE rev >= (SELECT percentile_cont(0.99) WITHIN GROUP (ORDER BY rev) FROM cust_rev)
ORDER BY rev DESC;

QUERY PLAN

Sort (cost=864.53..866.30 rows=708 width=40) (actual time=10.666..10.669 rows=18 loops=1)
  Sort Key: cust_rev.rev DESC
  Sort Method: quicksort Memory: 25kB
  Buffers: shared hit=257
  CTE cust_rev
    -> HashAggregate (cost=698.31..724.85 rows=2123 width=40) (actual time=9.441..9.791 rows=1707 loops=1)
      Group Key: o.customer_id
      Batches: 1 Memory Usage: 881kB
      Buffers: shared hit=248
      -> Hash Join (cost=249.42..659.67 rows=7729 width=14) (actual time=3.495..7.655 rows=7917 loops=1)
        Hash Cond: (oi.order_id = o.order_id)
        Buffers: shared hit=248
        -> Seq Scan on order_items oi (cost=0.00..361.93 rows=18393 width=14) (actual time=0.004..1.220 rows=18393 loops=1)
          Buffers: shared hit=178
        -> Hash (cost=213.86..213.86 rows=2845 width=16) (actual time=3.483..3.483 rows=2876 loops=1)
          Buckets: 4096 Batches: 1 Memory Usage: 167kB
          Buffers: shared hit=70
          -> Seq Scan on orders o (cost=0.00..213.86 rows=2845 width=16) (actual time=0.005..3.081 rows=2876 loops=1)
            Filter: ((order_status = ANY ('{paid,shipped,delivered}'::text[])) AND (order_ts >= (now() - '60 days'::interval)))
            Rows Removed by Filter: 3894
            Buffers: shared hit=70
  InitPlan 2
    -> Aggregate (cost=53.08..53.09 rows=1 width=8) (actual time=0.996..0.996 rows=1 loops=1)
      Buffers: shared hit=9
    -> CTE Scan on cust_rev cust_rev_1 (cost=0.00..42.46 rows=2123 width=32) (actual time=0.001..0.559 rows=1707 loops=1)
      -> CTE Scan on cust_rev (cost=0.00..53.08 rows=708 width=40) (actual time=10.462..10.659 rows=18 loops=1)
        Filter: ((rev)::double precision >= (InitPlan 2).col1)
        Rows Removed by Filter: 1689
        Buffers: shared hit=257
Planning:
  Buffers: shared hit=18
Planning Time: 0.197 ms
Execution Time: 10.711 ms
(33개 행)

skala_db=# 프
```

▲ 튜닝 전 — EXPLAIN (ANALYZE, BUFFERS) baseline 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x58

skala_db=# \i q10_after.sql
EXPLAIN (ANALYZE, BUFFERS)
WITH cust_rev AS (SELECT o.customer_id, sum(oi.line_total) AS rev
FROM orders o JOIN order_items oi ON oi.order_id=o.order_id
WHERE o.order_status IN ('paid','shipped','delivered')
AND o.order_ts >= now() - interval '60 days' GROUP BY o.customer_id)
SELECT customer_id, rev FROM cust_rev
WHERE rev >= (SELECT percentile_cont(0.99) WITHIN GROUP (ORDER BY rev) FROM cust_rev)
ORDER BY rev DESC;

QUERY PLAN

Sort (cost=843.46..845.23 rows=708 width=40) (actual time=9.246..9.249 rows=18 loops=1)
Sort Key: cust_rev.rev DESC
Sort Method: quicksort Memory: 25kB
Buffers: shared hit=258
CTE cust_rev
-> HashAggregate (cost=677.25..703.78 rows=2123 width=40) (actual time=7.771..8.298 rows=1707 loops=1)
Group Key: o.customer_id
Batches: 1 Memory Usage: 881kB
Buffers: shared hit=258
-> Hash Join (cost=228.36..638.60 rows=7729 width=14) (actual time=1.650..5.903 rows=7917 loops=1)
Hash Cond: (oi.order_id = o.order_id)
Buffers: shared hit=258
-> Seq Scan on order_items oi (cost=0.00..361.93 rows=18393 width=14) (actual time=0.005..1.265 rows=18393 loops=1)
Buffers: shared hit=178
-> Hash (cost=192.79..192.79 rows=2845 width=16) (actual time=1.633..1.634 rows=2876 loops=1)
Buckets: 4096 Batches: 1 Memory Usage: 167kB
Buffers: shared hit=80
-> Bitmap Heap Scan on orders o (cost=62.34..192.79 rows=2845 width=16) (actual time=0.249..1.313 rows=2876 loops=1)
Recheck Cond: ((order_ts >= (now() - '60 days'::interval)) AND (order_status = ANY ('{paid,shipped,delivered}'::text[])))
Heap Blocks: exact=70
Buffers: shared hit=80
-> Bitmap Index Scan on idx_orders_rev_ts (cost=0.00..61.62 rows=2845 width=0) (actual time=0.101..0.101 rows=2876 loops=1)
Index Cond: (order_ts >= (now() - '60 days'::interval))
Buffers: shared hit=10

InitPlan 2
-> Aggregate (cost=53.00..53.09 rows=1 width=8) (actual time=1.221..1.221 rows=1 loops=1)
-> CTE Scan on cust_rev cust_rev_1 (cost=0.00..42.46 rows=2123 width=32) (actual time=0.001..0.788 rows=1707 loops=1)
-> CTE Scan on cust_rev (cost=0.00..53.00 rows=708 width=40) (actual time=9.010..9.232 rows=18 loops=1)
Filter: ((rev)::double precision >= (InitPlan 2).col1)
Rows Removed by Filter: 1689
Buffers: shared hit=258

Planning:
Buffers: shared hit=12
Planning Time: 0.771 ms
Execution Time: 9.307 ms
(35% 행 )

skala_db=#
```

▲ 튜닝 후 — 인덱스/재작성 적용 후 실행계획

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x37

skala_db=# \i q10_result.sql
WITH cust_rev AS (SELECT o.customer_id, sum(oi.line_total) AS rev
FROM orders o JOIN order_items oi ON oi.order_id=o.order_id
WHERE o.order_status IN ('paid','shipped','delivered')
AND o.order_ts >= now() - interval '60 days' GROUP BY o.customer_id)
SELECT customer_id, rev FROM cust_rev
WHERE rev >= (SELECT percentile_cont(0.99) WITHIN GROUP (ORDER BY rev) FROM cust_rev)
ORDER BY rev DESC;
customer_id | rev
-----
19 | 33828.24
8 | 30856.78
9 | 28940.18
22 | 28859.38
24 | 26714.13
18 | 26244.07
10 | 24827.26
16 | 25131.91
4 | 24365.43
25 | 23359.85
20 | 22887.52
12 | 22727.50
2 | 22555.95
7 | 22298.63
6 | 22133.87
21 | 22049.44
26 | 21828.22
29 | 20696.36
(18개 행 )

skala_db=#
```

▲ 실행 결과 (튜닝 전후 동일 — 정확성 보존)

- 개선 — orders 접근이 Seq Scan → Bitmap Index Scan (idx_orders_rev_ts) (60일 ≈ 42%로 Q3보다 선택적이라 인덱스 채택).
- 결과 — 상위 1% = 18명 (고객 19번 33,828 ~ 29번 20,696, 전후 동일)
- 비즈니스 인사이트 — 최근 60일 기준 상위 1% = 18명이 고매출 구간을 형성. 소수 VIP의 최근 활성도가 높아 이탈 방지·전용 혜택의 ROI가 큼.

Q11. 안전 나눗셈 UDF(f_safe_div)로 채널별 AOV 쿼리 재작성

요구사항 — 분모 0이어도 에러 없이 0 반환하는 UDF로 채널별 AOV. 선택 컬럼: channel, revenue, orders, safe_aov.

병목 — orders = count(DISTINCT order_id) → GroupAggregate 입력 (channel, order_id) 14,250행 Sort.

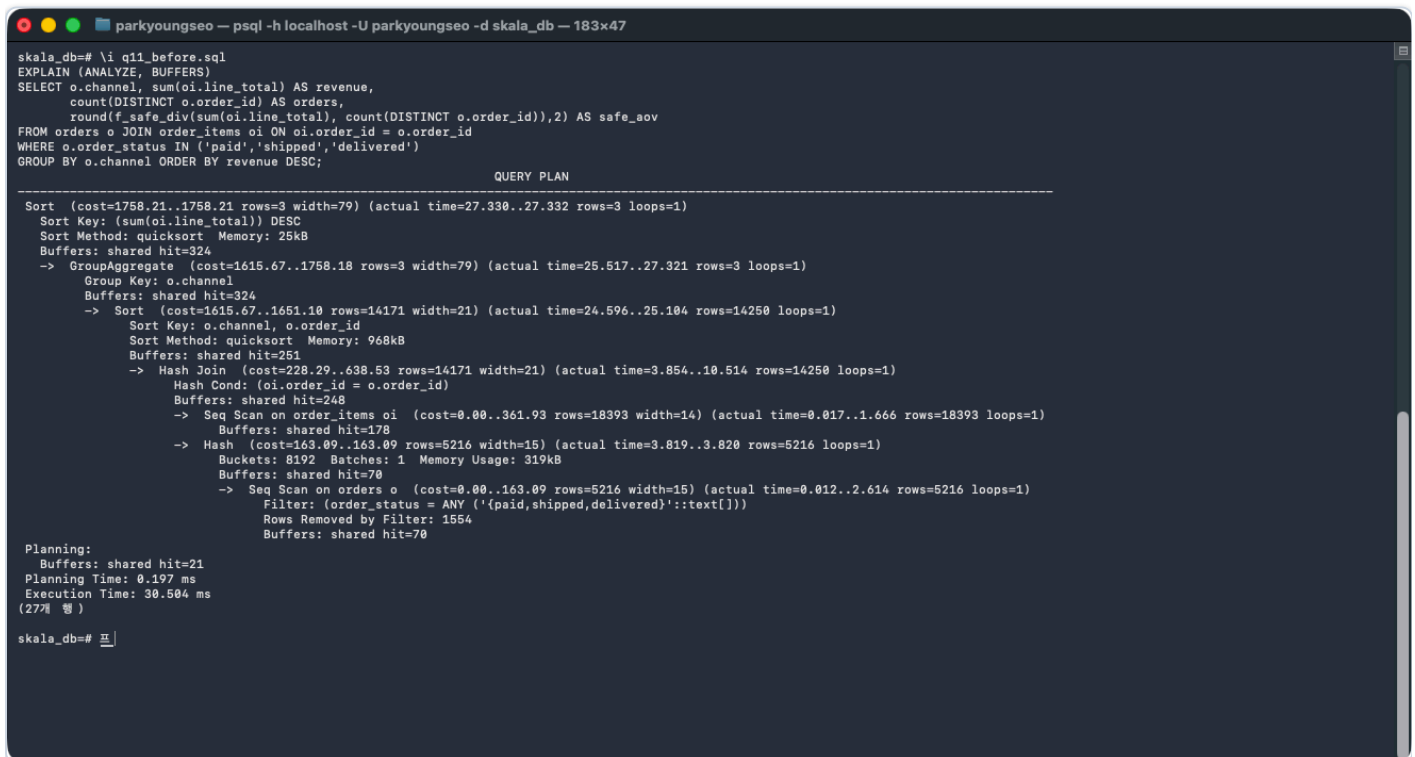
튜닝: 쿼리 재작성 — 주문 단위 선집계 후 채널별 재집계. f_safe_div로 분모 0 안전 처리

SQL — 튜닝 전 쿼리

```
SELECT o.channel, sum(oi.line_total) AS revenue,          -- 채널별 매출
       count(DISTINCT o.order_id) AS orders,             -- 주문수(DISTINCT)
       round(f_safe_div(sum(oi.line_total), count(DISTINCT o.order_id)),2) AS safe_aov -- 안전 AOV(0나눗셈 방지)
FROM   orders o
JOIN   order_items oi ON oi.order_id = o.order_id
WHERE  o.order_status IN ('paid','shipped','delivered')
GROUP BY o.channel ORDER BY revenue DESC;
```

SQL — 튜닝 후 (재작성 쿼리)

```
-- 주문 단위 선집계 후 채널별 재집계 (count(DISTINCT) 제거) + 안전 나눗셈 UDF
SELECT channel, sum(order_rev) AS revenue, count(*) AS orders, -- 채널별 재집계
       round(f_safe_div(sum(order_rev), count(*)),2) AS safe_aov -- 안전 AOV(주문수 0이면 0)
FROM (
  SELECT o.order_id, o.channel, sum(oi.line_total) AS order_rev -- 주문 단위 선집계
  FROM   orders o JOIN order_items oi ON oi.order_id = o.order_id
  WHERE  o.order_status IN ('paid','shipped','delivered')
  GROUP BY o.order_id, o.channel
) t
GROUP BY channel ORDER BY revenue DESC;
-- f_safe_div(100,0)=0 (에러 없음), f_safe_div(100,4)=25
```



```
parkyoungseo - psql -h localhost -U parkyoungseo -d skala_db - 183x47
scala_db=# \i q11_before.sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT o.channel, sum(oi.line_total) AS revenue,
       count(DISTINCT o.order_id) AS orders,
       round(f_safe_div(sum(oi.line_total), count(DISTINCT o.order_id)),2) AS safe_aov
FROM   orders o
JOIN   order_items oi ON oi.order_id = o.order_id
WHERE  o.order_status IN ('paid','shipped','delivered')
GROUP BY o.channel ORDER BY revenue DESC;

               QUERY PLAN

Sort  (cost=1758.21..1758.21 rows=3 width=79) (actual time=27.330..27.332 rows=3 loops=1)
  Sort Key: (sum(oi.line_total)) DESC
  Sort Method: quicksort  Memory: 25kB
  Buffers: shared hit=324
-> GroupAggregate  (cost=1615.67..1758.18 rows=3 width=79) (actual time=25.517..27.321 rows=3 loops=1)
  Group Key: o.channel
  Buffers: shared hit=324
-> Sort  (cost=1615.67..1651.10 rows=14171 width=21) (actual time=24.596..25.104 rows=14250 loops=1)
  Sort Key: o.channel, o.order_id
  Sort Method: quicksort  Memory: 968kB
  Buffers: shared hit=251
-> Hash Join  (cost=228.29..638.53 rows=14171 width=21) (actual time=3.854..10.514 rows=14250 loops=1)
  Hash Cond: (oi.order_id = o.order_id)
  Buffers: shared hit=248
-> Seq Scan on order_items oi  (cost=0.00..361.93 rows=18393 width=14) (actual time=0.017..1.666 rows=18393 loops=1)
  Buffers: shared hit=178
-> Hash  (cost=163.09..163.09 rows=5216 width=15) (actual time=3.819..3.820 rows=5216 loops=1)
  Buckets: 8192  Batches: 1  Memory Usage: 319kB
  Buffers: shared hit=70
-> Seq Scan on orders o  (cost=0.00..163.09 rows=5216 width=15) (actual time=0.012..2.614 rows=5216 loops=1)
  Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
  Rows Removed by Filter: 1554
  Buffers: shared hit=70

Planning:
  Buffers: shared hit=21
  Planning Time: 0.197 ms
  Execution Time: 30.504 ms
(27개 행)

scala_db=# 프
```

▲ 튜닝 전 — EXPLAIN (ANALYZE, BUFFERS) baseline 실행계획

제출물 2 · 3가지 조인 알고리즘 비교

동일 쿼리(최근 30일 주문 × order_items 집계)를 `enable_*` 플래그로 각 조인 전략을 강제하여 실행계획·특성을 비교한다. 외부 (orders) 1,836행 × 내부(order_items) 18,393행.

전략	실행계획 핵심	실행시간	유리한 상황
Nested Loop	외부 각 행마다 내부를 Index Only Scan으로 프로브 (loops=1,836, Heap Fetches 0)	약 7.6ms	외부가 작고 내부에 인덱스가 있을 때. 외부가 커지면 프로브 폭증에 불리.
Hash Join	orders로 해시 빌드 + order_items 전량 Seq Scan 후 매칭 (옵티마이저 기본 선택)	약 11.5ms	인덱스 없거나 대량 등가 조인에 안정적.
Merge Join	order_items는 인덱스 정렬 스캔, orders는 Sort 후 병합	약 7.8ms	양쪽이 이미 조인 키로 정렬되어 있을 때 매우 효율.

※ 위 실행시간은 각 캡처(강제 실행)의 EXPLAIN Execution Time이다. 단일 측정이라 편차가 커 알고리즘 우열의 근거가 못 된다 — 이 회차에선 강제 Hash가 11.5ms로 가장 높게 나왔지만 이는 계속 노이즈이며, 옵티마이저는 비용 추정상 Hash Join을 기본 선택한다(자동 선택 캡처). 핵심은 각 전략의 계획 구조와 적용 상황이다.

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x37

skala_db=# \i join0_auto.sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT o.order_id, sum(oi.line_total) AS order_total
FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
WHERE o.order_ts >= now() - interval '30 days'
GROUP BY o.order_id;

QUERY PLAN

HashAggregate (cost=646.64..669.60 rows=1837 width=40) (actual time=12.317..13.113 rows=1836 loops=1)
  Group Key: o.order_id
  Batches: 1 Memory Usage: 881kB
  Buffers: shared hit=248
  -> Hash Join (cost=211.44..621.68 rows=4991 width=14) (actual time=4.626..10.323 rows=5018 loops=1)
    Hash Cond: (oi.order_id = o.order_id)
    Buffers: shared hit=248
    -> Seq Scan on order_items oi (cost=0.00..361.93 rows=18393 width=14) (actual time=0.006..1.814 rows=18393 loops=1)
      Buffers: shared hit=178
    -> Hash (cost=188.48..188.48 rows=1837 width=8) (actual time=4.597..4.598 rows=1836 loops=1)
      Buckets: 2048 Batches: 1 Memory Usage: 88kB
      Buffers: shared hit=70
      -> Seq Scan on orders o (cost=0.00..188.48 rows=1837 width=8) (actual time=0.010..4.255 rows=1836 loops=1)
        Filter: (order_ts >= (now() - '30 days'::interval))
        Rows Removed by Filter: 4934
        Buffers: shared hit=70
Planning:
  Buffers: shared hit=12
Planning Time: 0.557 ms
Execution Time: 13.234 ms
(20개 행)

skala_db=#
```

▲ [기본] 옵티마이저 자동 선택 → Hash Join 채택 (자동 선택 데모)

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x41

skala_db=# \i join1_nlj.sql
SET enable_hashjoin=off; SET enable_mergejoin=off;
SET
SET
EXPLAIN (ANALYZE, BUFFERS)
SELECT o.order_id, sum(oi.line_total) AS order_total
FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
WHERE o.order_ts >= now() - interval '30 days'
GROUP BY o.order_id;

QUERY PLAN

HashAggregate (cost=1265.12..1288.08 rows=1837 width=40) (actual time=7.102..7.475 rows=1836 loops=1)
  Group Key: o.order_id
  Batches: 1 Memory Usage: 881kB
  Buffers: shared hit=3759
  -> Nested Loop (cost=0.29..1240.16 rows=4991 width=14) (actual time=0.045..5.882 rows=5018 loops=1)
    Buffers: shared hit=3759
    -> Seq Scan on orders o (cost=0.00..188.48 rows=1837 width=8) (actual time=0.025..1.867 rows=1836 loops=1)
      Filter: (order_ts >= (now() - '30 days'::interval))
      Rows Removed by Filter: 4934
      Buffers: shared hit=70
    -> Index Only Scan using idx_oi_order_prod_cov on order_items oi (cost=0.29..0.54 rows=3 width=14) (actual time=0.002..0.002 rows=3 loops=1836)
      Index Cond: (order_id = o.order_id)
      Heap Fetches: 0
      Buffers: shared hit=3689
Planning Time: 0.228 ms
Execution Time: 7.555 ms
(16개 행)

RESET enable_hashjoin; RESET enable_mergejoin;
RESET
RESET
skala_db=#
```

▲ [1] Nested Loop 강제 — Index Only Scan 프로브(loops=1,836)

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x43

skala_db=# \i join2_hash.sql
SET enable_nestloop=off; SET enable_mergejoin=off;
SET
SET
EXPLAIN (ANALYZE, BUFFERS)
SELECT o.order_id, sum(oi.line_total) AS order_total
FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
WHERE o.order_ts >= now() - interval '30 days'
GROUP BY o.order_id;

                                QUERY PLAN
-----
HashAggregate (cost=646.64..669.60 rows=1837 width=40) (actual time=10.769..11.343 rows=1836 loops=1)
  Group Key: o.order_id
  Batches: 1  Memory Usage: 881kB
  Buffers: shared hit=248
  -> Hash Join (cost=211.44..621.68 rows=4991 width=14) (actual time=3.977..9.072 rows=5018 loops=1)
    Hash Cond: (oi.order_id = o.order_id)
    Buffers: shared hit=248
    -> Seq Scan on order_items oi (cost=0.00..361.93 rows=18393 width=14) (actual time=0.008..1.710 rows=18393 loops=1)
      Buffers: shared hit=170
    -> Hash (cost=188.48..188.48 rows=1837 width=8) (actual time=3.946..3.946 rows=1836 loops=1)
      Buckets: 2048  Batches: 1  Memory Usage: 88kB
      Buffers: shared hit=70
      -> Seq Scan on orders o (cost=0.00..188.48 rows=1837 width=8) (actual time=0.010..3.517 rows=1836 loops=1)
        Filter: (order_ts >= (now() - '30 days'::interval))
        Rows Removed by Filter: 4934
        Buffers: shared hit=70
Planning Time: 0.166 ms
Execution Time: 11.485 ms
(18개 행)

RESET enable_nestloop; RESET enable_mergejoin;
RESET
RESET
skala_db=#
```

▲ [2] Hash Join 강제 — order_items 전량 스캔 + 해시

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x43

skala_db=# \i join3_merge.sql
SET enable_nestloop=off; SET enable_hashjoin=off;
SET
SET
EXPLAIN (ANALYZE, BUFFERS)
SELECT o.order_id, sum(oi.line_total) AS order_total
FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
WHERE o.order_ts >= now() - interval '30 days'
GROUP BY o.order_id;

                                QUERY PLAN
-----
GroupAggregate (cost=288.30..1050.12 rows=1836 width=40) (actual time=1.523..7.674 rows=1836 loops=1)
  Group Key: o.order_id
  Buffers: shared hit=285
  -> Merge Join (cost=288.30..1002.23 rows=4988 width=14) (actual time=1.493..6.399 rows=5018 loops=1)
    Merge Cond: (oi.order_id = o.order_id)
    Buffers: shared hit=285
    -> Index Scan using idx_order_items_order on order_items oi (cost=0.29..609.18 rows=18393 width=14) (actual time=0.008..2.711 rows=18378 loops=1)
      Buffers: shared hit=215
    -> Sort (cost=288.01..292.60 rows=1836 width=8) (actual time=1.474..1.569 rows=1836 loops=1)
      Sort Key: o.order_id
      Sort Method: quicksort  Memory: 49kB
      Buffers: shared hit=70
      -> Seq Scan on orders o (cost=0.00..188.48 rows=1836 width=8) (actual time=0.009..1.399 rows=1836 loops=1)
        Filter: (order_ts >= (now() - '30 days'::interval))
        Rows Removed by Filter: 4934
        Buffers: shared hit=70
Planning:
  Buffers: shared hit=282
Planning Time: 1.685 ms
Execution Time: 7.782 ms
(20개 행)

RESET enable_nestloop; RESET enable_hashjoin;
RESET
RESET
skala_db=#
```

▲ [3] Merge Join 강제 — Sort 후 병합

제출물 3 · Materialized View (mv_daily_gmv)

매일 총 판매금액(GMV) 리포트를 **매번 JOIN+SUM** 하면 느리므로, 사전 집계한 `mv_daily_gmv` 로 리포트 질의를 가속한다.

```
CREATE MATERIALIZED VIEW mv_daily_gmv AS
SELECT date_trunc('day', o.order_ts) AS day, sum(oi.line_total) AS gmv
FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
WHERE o.order_status IN ('paid','shipped','delivered')
GROUP BY 1;
CREATE UNIQUE INDEX ux_mv_daily_gmv_day ON mv_daily_gmv(day); -- CONCURRENTLY 갱신 대비
```

구분	실행계획	실행시간
MV 미사용 (JOIN+SUM 직접 집계)	Seq Scan×2 + Hash Join + HashAggregate (14,250행 집계)	약 19ms
MV 사용 (사전 집계 124행 조회)	Index Scan Backward (ux_mv_daily_gmv_day)	약 0.02ms

개선 — 캡처 EXPLAIN 기준 약 **860배** 단축(19ms→0.02ms). 세 제출물 중 **유일하게 대규모의 실질 시간 이득**으로, 집계 자체를 미리 계산해 두므로 `order_items` 전량 스캔·집계를 통째로 건너뛴다. 리포트성(반복) 질의에 Materialized View가 효과적. 비용: 저장공간 + 갱신(REFRESH) 필요. **갱신 전략**: 리포트가 오후 3시 기준이면 매일 15:00 스케줄러(cron/pg_cron)로 `REFRESH MATERIALIZED VIEW`; 무중단이 필요하면 `UNIQUE 인덱스 + REFRESH ... CONCURRENTLY`.

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x43

skala_db=# \i mv1_without.sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT date_trunc('day', o.order_ts) AS day, sum(oi.line_total) AS gmv
FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
WHERE o.order_status IN ('paid','shipped','delivered')
GROUP BY 1 ORDER BY 1 DESC LIMIT 10;

               QUERY PLAN
-----
Limit  (cost=935.77..935.80 rows=10 width=40) (actual time=18.898..18.902 rows=10 loops=1)
  Buffers: shared hit=248
  -> Sort  (cost=935.77..948.81 rows=5216 width=40) (actual time=18.897..18.899 rows=10 loops=1)
        Sort Key: (date_trunc('day'::text, o.order_ts)) DESC
        Sort Method: top-N heapsort  Memory: 26kB
        Buffers: shared hit=248
        -> HashAggregate  (cost=744.82..823.05 rows=5216 width=40) (actual time=18.818..18.879 rows=124 loops=1)
              Group Key: date_trunc('day'::text, o.order_ts)
              Batches: 1  Memory Usage: 273kB
              Buffers: shared hit=248
              -> Hash Join  (cost=228.29..673.96 rows=14171 width=14) (actual time=3.302..15.171 rows=14250 loops=1)
                    Hash Cond: (oi.order_id = o.order_id)
                    Buffers: shared hit=248
                    -> Seq Scan on order_items oi  (cost=0.00..361.93 rows=18393 width=14) (actual time=0.012..1.999 rows=18393 loops=1)
                          Buffers: shared hit=178
                    -> Hash  (cost=163.09..163.09 rows=5216 width=16) (actual time=3.255..3.256 rows=5216 loops=1)
                          Buckets: 8192  Batches: 1  Memory Usage: 309kB
                          Buffers: shared hit=70
                          -> Seq Scan on orders o  (cost=0.00..163.09 rows=5216 width=16) (actual time=0.009..2.271 rows=5216 loops=1)
                                Filter: (order_status = ANY ('{paid,shipped,delivered}'::text[]))
                                Rows Removed by Filter: 1554
                                Buffers: shared hit=70
Planning:
  Buffers: shared hit=18
Planning Time: 1.749 ms
Execution Time: 18.984 ms
(26개 행)

skala_db=#
```

▲ [MV 미사용] 일자별 GMV 직접 집계 — JOIN+SUM (Hash Join + HashAggregate), Execution 약 19ms

```
parkyoungseo — psql -h localhost -U parkyoungseo -d skala_db — 183x22

skala_db=# \i mv2_with.sql
REFRESH MATERIALIZED VIEW mv_daily_gmv;
REFRESH MATERIALIZED VIEW
EXPLAIN (ANALYZE, BUFFERS)
SELECT day, gmv FROM mv_daily_gmv ORDER BY day DESC LIMIT 10;

               QUERY PLAN
-----
Limit  (cost=0.14..1.26 rows=10 width=16) (actual time=0.010..0.014 rows=10 loops=1)
  Buffers: shared hit=1 read=1
  -> Index Scan Backward using ux_mv_daily_gmv_day on mv_daily_gmv  (cost=0.14..14.00 rows=124 width=16) (actual time=0.009..0.012 rows=10 loops=1)
        Buffers: shared hit=1 read=1
Planning:
  Buffers: shared hit=21 read=2
Planning Time: 0.210 ms
Execution Time: 0.022 ms
(8개 행)

skala_db=#
```

▲ [MV 사용] REFRESH 후 같은 리포트를 MV에서 조회 — Index Scan Backward, Execution 약 0.02ms

```
parkyoungseo - psql -h localhost -U parkyoungseo -d skala_db - 183x28
skala_db=# \i mv3_result.sql
SELECT day::date, gmV FROM mv_daily_gmv ORDER BY day DESC LIMIT 10;
 day | gmV
-----+-----
2026-08-08 | 700.10
2026-07-28 | 260.26
2026-07-26 | 337.14
2026-07-24 | 27889.06
2026-07-23 | 35766.44
2026-07-22 | 61289.70
2026-07-21 | 45305.23
2026-07-20 | 57629.43
2026-07-19 | 43508.58
2026-07-18 | 37398.53
(10개 행 )

SELECT count(*) AS mv_rows FROM mv_daily_gmv;
 mv_rows
-----
124
(1개 행 )

skala_db=#
```

▲ MV 결과 미리보기 (일자별 GMV) 및 행 수(124일)

종합 요약

#	튜닝 기법	계획 변화 (전 → 후) — 핵심 근거	EXPLAIN 실행시간 (캡처, 전→후)
Q1	부분 인덱스	Seq Scan → Bitmap Index Scan (orders 접근 인덱스화)	8.6 → 7.5ms
Q2	쿼리 재작성	GroupAggregate+Sort(14,250행) → HashAggregate	21.1 → 19.8ms
Q3	계획 분석	Seq Scan 유지 (90일 저선택도, 옵티마이저 판단 정당)	12.7 → 13.9ms (계획 동일)
Q4	계획 분석	전량 집계·RANK Sort 불가피 → 전처리(MV)가 정당	14.2 → 18.5ms (계획 동일)
Q5	쿼리 재작성	GroupAggregate+Sort(14,250행) → HashAggregate	14.2 → 18.3ms *
Q6	복합 부분 인덱스	Index Scan(heap 재확인, buf 5,539) → Index Only Scan	6.0 → 10.0ms * (버퍼 5,539→4,460)
Q7	부분 인덱스	Seq Scan(600) → Bitmap Index Scan(61) — 접근행수↓	0.21 → 0.17ms *
Q8	커버링 인덱스	Seq Scan → Index Only Scan (Heap Fetches 0)	2.3 → 1.5ms
Q9	쿼리 재작성	GroupAggregate+Sort(5,216행) → HashAggregate	21.8 → 18.0ms
Q10	부분 인덱스	Seq Scan → Bitmap Index Scan (60일~42%)	10.7 → 9.3ms
Q11	쿼리 재작성+UDF	이중 Sort(14,250행+외부) → HashAggregate	30.5 → 15.4ms
조인	NLJ/Hash/Merge	전략별 실행계획·특성 비교	NLJ 7.6·Hash 11.5·Merge 7.8ms
MV	사전 집계	JOIN+SUM(전량 집계) → Index Scan (MV 124행)	19 → 0.02ms (~860배)

※ **측정 유의점** — 데이터 규모(orders 6,770·order_items 18,393)가 작아 절대 실행시간은 수 ms대이고, EXPLAIN ANALYZE의 행별 계속 오버헤드로 **단일 실행 시간은 실행마다 편차가 크다**. 위 표의 시간은 각 문항 캡처(EXPLAIN ANALYZE)의 **Execution Time 전→후 값**이다. 이 편차 때문에 *표시 문항은 튜닝 후 단일 측정을 곧이곧대로 읽으면 안 된다 — Q5·Q6은 후가 오히려 근소히 높게 나왔지만(노이즈) 계획·버퍼는 개선(Q5 대형 Sort 제거, Q6 버퍼 5,539→4,460·Heap Fetches 0), Q7은 강제 인덱스(후)가 근소히 빨랐으나 옵티마이저는 비용상 Seq Scan을 유지(접근 행수 600→61). 따라서 튜닝의 핵심 근거는 시간이 아니라 **실행계획 노드(스캔·조인·정렬)와 접근 행수·버퍼의 변화**이며, 이 효과는 데이터가 커질수록 실질 이득으로 확대된다. **인덱스가 항상 이득은 아니며**(Q3·Q4는 Seq Scan 유지가 최적, Q7은 소형 테이블이라 인덱스 이득이 측정 노이즈 수준 — 옵티마이저는 비용상 Seq Scan 유지), 비용기반 옵티마이저의 선택을 함께 분석하였다. 실질적으로 큰 시간 이득은 **Materialized View(약 860배)**에서 나온다.

스캔·인덱스·조인 전략 정리

- **스캔 방식** — Seq Scan(소형·저선택도 필터: Q3·Q4·Q7·Q8에서 옵티마이저 채택), **Bitmap Index Scan**(부분 인덱스로 선택적 구간만: Q1·Q10), **Index Only Scan**(커버링/복합 인덱스로 heap 접근 제거, Heap Fetches 0: Q6·Q8)를 상황별로 확인.
- **인덱스 전략** — **부분 인덱스**(매출 상태·품질 조건만 색인: Q1·Q7·Q10), **복합 부분 인덱스**((customer_id, order_ts)+status: Q6), **커버링 인덱스**(product_id INCLUDE rating: Q8). 인덱스 크기·유지비를 낮추면서 필요한 행만 겨냥.
- **쿼리 재작성** — count(DISTINCT)가 유발하는 대형 Sort를 **주문 단위 선집계** → HashAggregate로 제거(Q2·Q5·Q9·Q11). 인덱스 없이 계획 자체를 개선.
- **조인 전략** — 이 데이터에선 대량 등가 조인에 **Hash Join**이 기본 선택. 외부가 작고 내부에 인덱스가 있으면 **Nested Loop**(Index Only Scan 프로브), 정렬된 입력이면 **Merge Join**이 유리(제출물 2 참조).
- **사전 집계** — 반복 조회되는 리포트성 질의는 **Materialized View**로 집계를 미리 계산해 전량 스캔·집계를 통째로 회피(제출물 3, 약 860배).

부록 · DB 엔진별 옵티마이저 특징 비교

같은 SQL이라도 엔진마다 옵티마이저 동작·튜닝 접근이 다르다. 본 실습(PostgreSQL) 관점에서 주요 엔진을 정리한다.

엔진	옵티마이저 특징	튜닝 관점
PostgreSQL (본 실습)	통계 기반의 엄격한 비용기반 옵티마이저(CBO) . Hash Join·Bitmap Heap Scan·부분/커버링 인덱스 등 분석(OLAP)에 강점.	EXPLAIN (ANALYZE, BUFFERS) 로 계획·버퍼 분석, 부분/커버링 인덱스·Materialized View 적극 활용.
MySQL	전통적으로 OLTP 중심이라 Nested Loop 의존이 컸으나 8.0부터 Hash Join 지원으로 대량 집계 성능 개선.	커버링 인덱스·복합 인덱스 설계가 핵심. 옵티마이저 힌트 제한적.
Oracle	가장 방대·정교한 통계와 튜닝 힌트(Hint) . DBA가 실행계획을 세밀하게 제어 가능(상용 강자).	Hint·SQL Plan Baseline으로 계획 고정, 파티셔닝·병렬 활용.
SQL Server	고도로 자동화된 옵티마이저. 실행계획 분석 시 Missing Index(누락 인덱스) 추천 , CPU 병렬 처리에 강점.	추천 인덱스 검토 + 실행계획 캐시(Plan Cache) 관리.

※ 공통 원리는 동일하다 — **필요한 행만 읽고(스캔·인덱스)**, **불필요한 정렬을 줄이며(재작성)**, **반복 집계는 미리 계산(MV)**. 엔진별 차이는 '어떤 계획을 어떻게 강제·유도하느냐'의 방법론 차이다.